

SpineXR-VQA: A Clinically-Validated Visual Question Answering Dataset for Spine X-Rays

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Recent progress in Medical Visual Question Answering (VQA) has significantly aided clinical decision-making across various domains such as pneumonia, oncology, and neurology. However, spinal and musculoskeletal ailments remain critically under-explored. The development of reliable VQA models for spinal imaging is currently hindered by a lack of datasets and evaluation protocols that fail to mirror the descriptive, diagnostic interpretations used in actual clinical practice. In this resource paper, we introduce **SpineXR-VQA**, an open-source, clinically grounded and verified benchmark comprising 2,187 X-rays and 8,272 expert-verified, open-ended Question-Answer pairs. Unlike standard classification-based datasets, **SpineXR-VQA** features six expert-validated categories: abnormality, severity, location, diagnosis, treatment, and reasoning. Ten orthopedic specialists from India and Thailand validated all pairs, ensuring geographic diversity and high inter-rater agreement (Cohen’s Kappa: 0.96 for questions, 0.93 for answers). We benchmark fifteen state-of-the-art Multimodal Large Language Models (MLLMs), including proprietary systems like Claude 3.5 Sonnet, GPT-4o, and Gemini 1.5 Flash, alongside medical-specific and general open-weight models. While these models achieve moderate semantic alignment (median similarity: 0.72), detailed analysis reveals they consistently fail to capture essential diagnostic nuances like anatomical fidelity and clinical completeness. These results underscore the necessity for specialized models in spinal VQA, a gap **SpineXR-VQA** is designed to fill.

1 Introduction

According to the Global Burden of Disease Study [12, 41] degenerative spinal diseases are among the most prevalent musculoskeletal disorders worldwide, affecting more than 540 million people and accounting for nearly 10% of all years lived with disability (YLDs). Radiological assessment of the spine remains the cornerstone of diagnosis and treatment planning [8, 29], yet the interpretive process is inherently complex: subtle variations in vertebral alignment, disc height, cortical margins, and soft-tissue shadows can fundamentally influence clinical conclusions [34]. This diagnostic difficulty is compounded by high global demand for spine imaging, persistent radiologist shortages, and well-documented inter-observer variability—all of which motivate the development of reliable automated assistance tools.

Recent advances in Multimodal Large Language Models (MLLMs) [21, 30, 31] have created optimism that such systems could assist clinicians by jointly interpreting medical images and text. Their ability to integrate visual context with natural-language reasoning makes them particularly appealing for tasks such as medical visual question answering (MedVQA) [10, 11, 43]. However, deploying these systems in spinal imaging remains

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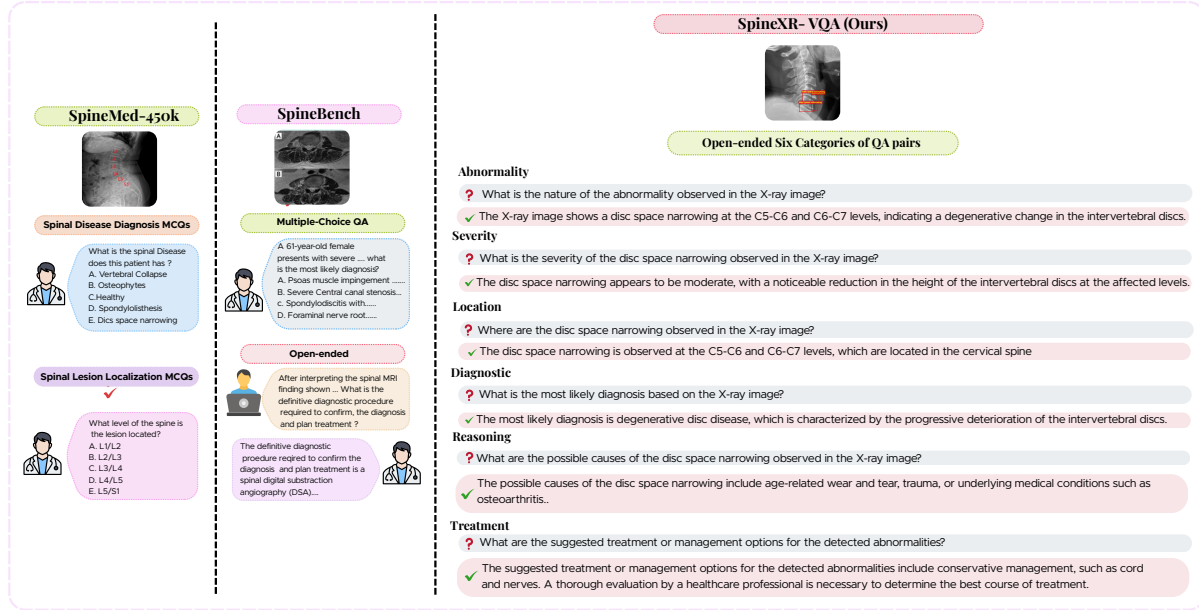


Fig. 1. Comparative overview of question-answer design across existing spine VQA datasets and the proposed **SpineXR-VQA**. Unlike prior semi-automatically generated corpora, **SpineXR-VQA** includes fully clinician-validated, open-ended questions spanning six diagnostic categories.

challenging. Despite rapid progress in model architectures, the field still lacks datasets and benchmarks that satisfy three prerequisites for clinically meaningful VQA for spine imaging: (1) clinically validated annotations, (2) precise anatomical localization, and (3) reasoning-oriented, open-ended QA pairs. Without these prerequisites, automated systems lack the evaluative foundation necessary for clinical reliability; consequently, models trained on such deficient data will struggle significantly to generate clinically sound answers that meet the criteria of real-world diagnostic settings.

Although several MedVQA benchmarks have emerged, most prioritize breadth across imaging modalities rather than depth within specific anatomical regions. Datasets such as VQA-RAD [20], SLAKE [26], PathVQA [14], DME[39] and OmniMedVQA [16] encompass radiology, pathology, ophthalmology, and beyond. While valuable for broad anatomical coverage VQA research, their reliance on semi-automatically generated QA pairs (that are not clinically verified) and modality-agnostic annotations limits their utility for spine-specific diagnostic tasks. These datasets do not capture the nuanced anatomy, degenerative patterns, or reasoning processes required for interpreting spinal radiographs.

To address these limitations, recent spine-focused datasets have made important progress, yet they still fall short of supporting clinically reliable visual question answering. SpineMed-450K [49] contains over 55k VQA pairs generated via a clinician-in-the-loop prompting pipeline; however, only 0.1% of the data is manually reviewed by orthopedic surgeons. SpineBench [45] offers 64,878 automatically synthesized multiple-choice questions with limited clinician involvement. While these resources enable large-scale benchmarking, they remain predominantly synthetic, sparsely verified, and heavily biased toward classification-style evaluation (Figure 1). Similarly, although SpineMed-450K includes a substantial number of open-ended (35k) and closed-ended (20k) QA pairs, its open-ended questions are neither organized by clinical reasoning type nor manually validated, limiting their ability to

model the multi-sentence, explanatory reasoning characteristic of real-world radiology. Together, these gaps underscore the need for a clinically grounded, expert-verified, open-ended VQA dataset for spine related ailments.

In this work, we introduce **SpineXR-VQA**, a clinically-verified benchmark explicitly designed to address the limitations of prior spinal VQA datasets. **SpineXR-VQA** contains 2, 187 spine X-rays and 8, 272 QA pairs, each authored or reviewed by orthopedic specialists, radiologists, and spine surgeons from India and Thailand (five experts from each country). A subset was cross-validated between the two clinical teams to compare diagnostic phrasing, interpretive styles and inter-country agreement. Unlike semi-automatically generated corpora, **SpineXR-VQA** consists entirely of expert-validated, open-ended questions, spanning six clinically grounded reasoning categories: abnormality detection, severity assessment, anatomical location, diagnosis, treatment, and reasoning.

Although smaller than large synthetic datasets, **SpineXR-VQA** emphasizes quality over quantity, prioritizing interpretability, clinical fidelity, and cross-institutional reliability, attributes essential for real-world deployment. We conduct a comprehensive baseline evaluation across 15 state-of-the-art MLLMs, including proprietary, general-purpose, and medical-specialized models. Despite their scale, current models perform poorly on generative spinal VQA. Although they achieve moderate semantic similarity (median 0.72), they fail to preserve anatomical fidelity and clinical completeness, reflecting three systemic weaknesses: (1) limited adaptation to the visual and conceptual demands of radiographic imaging; (2) an inherent bias toward closed-ended, classification-style VQA rather than generative clinical reasoning; and (3) insufficient exposure to clinically grounded, open-ended question-answer distributions. These findings underscore the need for models that are explicitly engineered for spine imaging and are capable of generating clinically coherent diagnostic answers. It further highlights the inadequacy of standard NLP metrics for evaluating medical VQA, motivating the development of domain-specific, clinically meaningful evaluation protocols. In this resource paper, we address the above limitations and make the following contributions:

- (1) We introduce **SpineXR-VQA**, the first fully clinician-verified, open-ended, open-source spine X-rays VQA benchmark, comprising of 8, 272 expert-validated pairs across six clinically grounded reasoning categories.
- (2) We benchmark, state-of-the-art MLLMs on the challenging **SpineXR-VQA** pointing towards challenges with the existing models and pointing towards development of domain-adapted multimodal models for spinal VQA.
- (3) **(We release the code and supplementary material via an anonymous GitHub repository: <https://anonymous.4open.science/r/SpineXR-VQA-5E6D>. Upon acceptance, the dataset will be publicly released on Hugging Face.)***Deepali–edit (R2–comment)Deepali–Comment*

The paper is structured as follows. Section 2 surveys related work in medical visual question answering. Section 3 introduces the proposed methodology, while Section 4 describes the dataset curation pipeline. Section 5 provides a comprehensive analysis of the **SpineXR-VQA** dataset, followed by model benchmarking and ablation study in Section 6. Section 7 discusses the implications of our findings and outlines future research directions. Section 8 addresses the limitations of this study, Ethical Considerations for this dataset is discussed in Section 9, and Section 10 concludes the paper.

2 Related Work

2.1 Medical Visual Question Answering (MedVQA)

Medical Visual Question Answering (MedVQA) [1, 6, 7, 14, 20, 27] focuses on enabling models to interpret medical images and respond to natural-language queries within a clinical context. Early work such as VQA-RAD [20] established the feasibility of MedVQA through expert-annotated radiology data. Subsequent datasets broadened the scope of MedVQA across imaging modalities and anatomical regions, including PathVQA [14] for histopathology and SLAKE [26] for multi-organ CT and MRI imaging. Additional benchmarks, such as VQA-Med [1, 6, 7], OVQA [17], and DME-VQA, further extended MedVQA to extremities, ophthalmology, and

multi-organ settings, but largely relied close-ended questions limiting their alignment with real-world clinical reasoning.

Recent MedVQA benchmarks prioritize scale through automatic or semi-automatic question generation. Datasets such as MIMIC-CXR VQA [4], Medical-Diff-VQA [15], and PMC-VQA [48] generate large volumes of QA pairs from reports or captions, while OmniMedVQA [16] synthesizes approximately 127,995 pairs across 12 modalities and 20 anatomical regions using vision–language models. Although these resources enable large-scale benchmarking, their reliance on templated or caption-driven synthesis with limited physician oversight restricts anatomical fidelity and diagnostic realism.

Critically, existing MedVQA datasets provide limited spine-focused supervision and largely lack spine-specific, open-ended generative QA. Where spine-related questions are present, they are typically sparse or framed as closed-form classification, rendering them insufficient for learning the diagnostic reasoning required for spinal pathology assessment. A structured comparison of existing MedVQA datasets across modality, anatomical coverage, QA construction, and scale is provided in Table 1, highlighting the absence of clinically grounded, spine-specific generative benchmarks and motivating the need for targeted dataset development.

2.2 Spine-Specific Benchmarks

Within the spinal imaging domain, several datasets have been developed for tasks such as vertebral labeling, fracture detection, and report generation; however, most are not designed to support visual question answering or open-ended clinical reasoning. For example, datasets such as BUU-LSpine [18], RSNA Spine [25], VinDr-SpineXR [34], and CSXA [36] provide annotations for vertebral localization, pathology classification, or structured reporting from X-ray or CT images. While these resources are valuable for image classification or segmentation driven tasks, they do not include question–answer supervision and therefore cannot evaluate or train models for medical visual question answering.

The most relevant spine-specific VQA benchmarks to date are SpineMed-450K [49] and SpineBench [45]. SpineMed-450K contains over 460,000 QA pairs generated via a clinician-in-the-loop pipeline, yet only approximately 574 samples (0.1%) are manually verified by experts. SpineBench introduces 64,878 multiple-choice QA pairs synthesized using Gemini-2.5-Pro with SigLIP2-based hard-negative sampling, of which 2,128 samples undergo manual validation. Although these benchmarks advance large-scale evaluation, their QA construction remains predominantly machine-generated and classification-oriented, offering limited support for spine-specific, open-ended generative reasoning or clinically contextual inference.

2.3 Comparison with Other Datasets

As summarized in Table 2, **SpineXR-VQA** differs from existing spine-specific VQA benchmarks by emphasizing full clinical verification and open-ended, generative reasoning. All 8,272 QA pairs are clinically validated by practicing clinicians from India and Thailand and span six diagnostic reasoning categories aligned with routine radiology workflows.

In contrast, SpineMed-450K and SpineBench prioritize scale through predominantly LLM generated QA pairs with limited expert validation and a strong bias toward closed-form or classification-style questions. Consequently, these benchmarks provide limited support for evaluating clinically grounded, generative reasoning. By design, **SpineXR-VQA** serves both as a reliable benchmark for interpretability-focused evaluation and as a foundation for developing domain-adapted MLLMs aligned with clinical spine radiology.

2.4 Potential Biases in **SpineXR-VQA**

A potential limitation of **SpineXR-VQA** is that expert validation was conducted by clinicians from India and Thailand, which could introduce regional practice-pattern bias. However, this focus is also clinically grounded: spinal

Table 1. Descriptive comparison of existing MedVQA datasets across modalities, anatomical coverage, QA curation, no. of images and Qa Pairs.

Dataset	Year	Modality	Anatomy Covered	QA Creation	# Images	# QA Pairs
VQA-Med[13]	2018	X-ray	Abdomen, Head, Other	Semi-automatic	2,886	6,413
VQA-RAD[20]	2018	X-ray	Head, Chest, Abdomen	Manual	315	3,515
RadVisDial[19]	2019	X-ray	Chest	Automatic	91,060	455,300
RadVisDial (Gold)[19]	2019	X-ray	Chest	Manual	100	500
VQA-Med [6]	2019	X-ray/CT/MRI	Multi-organ	Semi-automatic	4,200	15,292
VQA-Med[1]	2020	X-ray/CT/MRI/Ultrasound	Multi-organ	Semi-automatic	5,000	5,000
SLAKE[26]	2021	CT/MRI	Neck, Chest, Abdomen, Pelvic, Head	Manual	642	14,000
VQA-Med [7]	2021	X-ray	Chest	Semi-automatic	5,000	5,000
OVQA [17]	2022	X-ray	Chest, Head, Leg, Hand	Semi-automatic	2,001	19,020
DME-VQA [39]	2022	Fundus	Retina (Mac-ula)	Manual	1,200	6,000
Medical-Diff-VQA [15]	2023	X-ray	Chest	Semi-automatic	16,432	700,703
MIMIC-CXR VQA [4]	2023	X-ray	Chest	Automatic	142,279	377,391
PMC-VQA [48]	2023	Mixture	Multiple	Automatic	149,000	227,000
PathVQA2.0 [14]	2024	Histopathology	Multiple or-gans	Semi-automatic	~60,000	~60,000
OmniMedVQA[16]	2024	12 modalities	20 anatomical regions	Automatic	118,010	127,995
SpineXR-VQA(Ours)	2025	X-rays	Spine	Manual	2,187	8,272

disorders represent a substantial public health burden, particularly in low- and middle-income regions/countries. Globally, over 15 million people live with spinal cord injury¹, with hundreds of thousands of new cases occurring annually due to trauma, falls, and road traffic accidents. In India alone, it has been estimated that approximately 1.5 million people are living with spinal cord injury, with around 20,000 new cases each year². While precise national estimates for Thailand are limited, musculoskeletal and spinal conditions such as low back pain and trauma are highly prevalent, with one regional study reporting low back pain prevalence of 26.7 % among adolescents³ and spinal trauma incidence among general trauma patients at 6 % in other developing world contexts. These figures reflect the large clinical exposure that spine specialists in India and Thailand have to a wide range of spinal pathologies, enhancing the relevance of clinician-verified QA pairs in **SpineXR-VQA**. Gathering knowledge and

¹<https://www.who.int/news-room/fact-sheets/detail/spinal-cord-injury>

²<https://www.neomotion.in/post/spinal-cord-injury-statistics-2023-2025-global-india-usa-uk-data-solutions>

³<http://ccsenet.org/journal/index.php/gjhs/article/view/71731>

Table 2. Comparison of spine-specific VQA benchmarks in terms of scale, clinical verification, QA design, and diagnostic coverage.

Dataset	Images	QA Pairs	Clinically Verified	Question Type	QA Source
SpineMed-450K [49]	33,123	55,248	Partial (~0.1%)	Open + Closed	LLM-generated (clinician-in-the-loop)
SpineBench [45]	40,263	64,878	Partial (~3%)	Closed (MCQ)	LLM-generated+hard negatives
SpineXR-VQA (Ours)	2,187	8,272	Yes (100%)	Open-ended	LLM-generated+clinician-validated

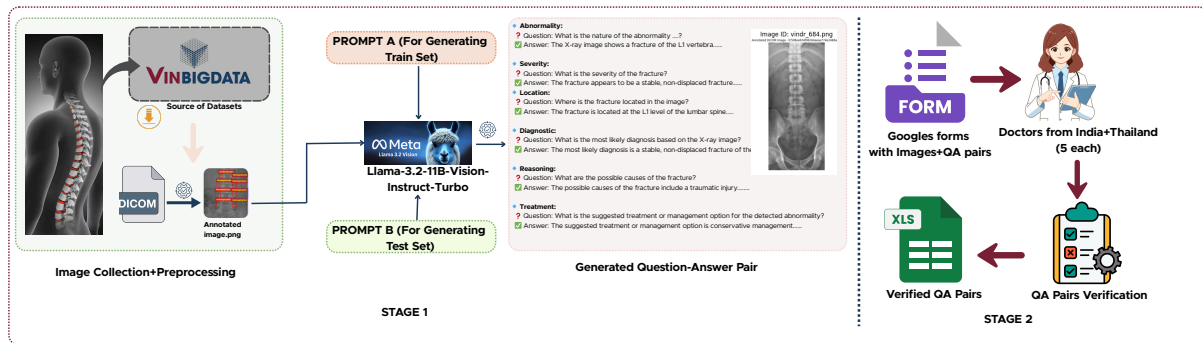


Fig. 2. Overview of the **SpineXR-VQA** curation pipeline. Stage 1: semi-automatic QA generation using LLaMA 3.2 Vision-Instruct with distinct prompts for train/test sets. Stage 2: expert verification by 10 clinicians (India + Thailand) using structured Google Forms.

experience from medical experts is tedious and time consuming process, nonetheless, in future work we plan to incorporate experts from additional regions to further improve generalizability and mitigate geographic bias.

3 Methodology

The development of **SpineXR-VQA** originated from a joint Indo–Thai joint research initiative aimed at addressing the lack of clinically grounded, anatomy-specific visual question answering (VQA) models and datasets for musculoskeletal diseases. Insights from an earlier survey on the *Barriers to integrating medical VQA systems into clinical practice* [28] informed the design of this dataset. The study, conducted across major hospitals and medical universities in **India** and **Thailand**, highlighted a consistent preference among clinicians for anatomy- and modality-specific datasets, given that diagnostic interpretation protocols differ substantially across anatomical regions and imaging modalities. Participants also emphasized that open-ended, generative question–answer formats better capture the descriptive and interpretive reasoning used in clinical radiology than multiple-choice or close-ended templates. Based on the above mentioned insights and after consultations with doctors from reputed hospitals from India and Thailand **spinal region** was selected.

It is important to note that, to avoid any potential biases, no patient images or proprietary clinical data were sourced from the hospitals whose doctors were involved in this research ; the hospitals participated exclusively in the *clinical design and verification* of the dataset. Their role included defining clinically meaningful QA categories and validating the correctness of the generated question–answer pairs. Through a series of consensus meetings, six reasoning-based QA categories were finalized—covering abnormality, severity, location, diagnostic, treatment and reasoning. Prior to dataset curation, **Institutional Review Board (IRB)** approval was obtained with Reference No.(**RERC 2024/054**). This collaborative and ethically grounded framework ensured that **SpineXR-VQA** was developed as a clinically relevant benchmark emphasizing interpretability, reasoning fidelity, and real-world diagnostic applicability.

4 Dataset Curation

The dataset was curated in two major stages, as illustrated in Figure 2. **Stage 1** focused on image selection, preprocessing, and prompt-based question–answer generation and **Stage 2** involved clinical validation, quality refinement, and the final data consolidation process (Figure 2).

4.1 Stage 1: Dataset Creation

In Stage 1, we constructed high-quality visual question–answer pairs from spine x-rays via standardized preprocessing and prompt-based generation. This approach ensures reproducibility and data integrity, generates clinically meaningful questions, and establishes a robust foundation for subsequent Stage 2 expert validation.

4.1.1 Stage 1.1: Image Selection and Preprocessing. For image selection, we utilized the publicly available **VinDr-SpineXR** dataset [34], which consists of spine radiographs annotated with 13 types of spinal abnormalities, including osteophytes, vertebral collapse, disc-space narrowing, and spondylolisthesis. All DICOM files were converted into high-resolution PNG format to facilitate compatibility with vision–language model (VLM) pipelines. **(Corresponding annotation overlays were then mapped onto the PNGs to provide interpretable ground-truth locations of abnormalities—an essential component for evaluating model attention and reasoning. These overlays were used solely during the dataset generation process and were not utilized during model fine-tuning or inference in the experiments described in Section 6.)**^{Deepali–edit (R3-comment3)}^{Deepali–Comment} Images for the **training** and **test** sets were selected according to the official VinDr train–test split. After filtering for quality and anatomical completeness, a total of **2,187 images** were retained, comprising both normal (54.0%) and abnormal (46.0%) cases. To ensure consistent formatting, each image was renamed systematically (e.g., `vindr_train_001.png`, `vindr_test_001.png`) for reproducibility and standardized dataset organization. We utilized the VinDr-SpineXR dataset to circumvent potential bias issues, ethical and legal barriers associated with primary clinical data acquisition from hospitals in India and Thailand. Unlike hospital-acquired data, which is often restricted by IRB constraints and consent requirements, VinDr-SpineXR provides a de-identified, expert-annotated, and standardized framework. This choice ensures cross-study comparability, avoids site-specific imaging biases, and allows for the free redistribution of the resulting dataset to the research community.

4.1.2 Stage 1.2: Prompt-Based QA Pair Generation. Once the preprocessed images were finalized, they were fed into two vision–language models—a closed-weights model, **Gemini 2.5 Pro**, and an open-weights model, **LLaMA 3.2 11B Vision-Instruct**. While there are several options available for vision-language models such as GPT5, Qwen, etc., based on our initial study involving several models and costs considerations we selected one model from close-weight category (**Gemini 2.5 Pro**) and one from the open-weight category (LLaMA 3.2 11B Vision-Instruct). Both models were prompted **((Figure 9 in Appendix A))**^{Deepali–edit} to generate synthetic question–answer pairs in English. To assess cross-model reliability of the generated QA pairs, both

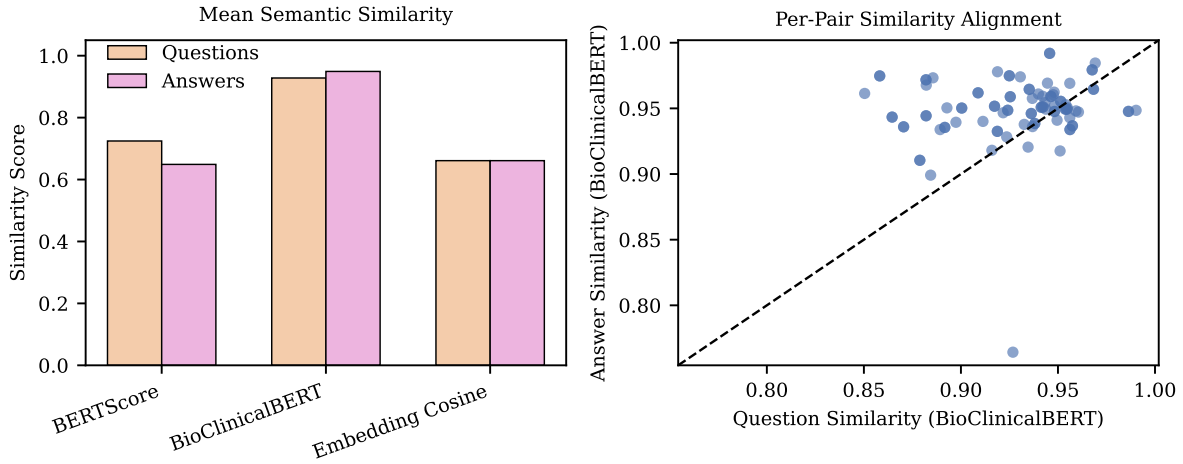


Fig. 3. Semantic similarity between QA pairs generated by LLaMA 3.2 11B Vision-Instruct and Gemini 2.5 Pro. The bar chart reports mean similarity across BERTScore, BioClinicalBERT cosine, and embedding cosine metrics. The scatter plot shows per-pair BioClinicalBERT similarity for questions (x-axis) and answers (y-axis), with the dashed line indicating $y = x$. High semantic agreement (BioClinicalBERT > 0.9) demonstrates that both models yield comparable radiology-focused QA reasoning, motivating the use of LLaMA for large-scale dataset generation.

models were applied to an initial subset of 96 images. We then computed lexical (BERTScore) and semantic (BioClinicalBERT and embedding-based cosine) similarity between the LLaMA- and Gemini-generated question-answer pairs (Figure 3). The resulting scores showed high semantic agreement between the two models (e.g., $\text{BERTScore}_Q = 0.72$, $\text{BERTScore}_A = 0.65$, BioClinicalBERT cosine > 0.92 for both questions and answers), indicating that Gemini and LLaMA induce very similar radiology-focused reasoning. Given this comparable output quality, we selected **LLaMA 3.2 11B Vision-Instruct** as the primary generator due to its open-source license, substantially lower inference cost, and compatibility with on-premise/offline clinical deployments. Some QA samples generated by both the models are shown in (Appendix E Table 15.)^{Deepali-edit}

As shown in Figure 2, two distinct prompts were employed: **Prompt A** for generating the training set and **Prompt B** for the test set. This ensured that train-test pairs were **lexically different but semantically equivalent**, minimizing (Lexical overfitting)^{Deepali-edit} during model fine-tuning while preserving consistent clinical reasoning. Each abnormal image produced six QA pairs covering the categories of *abnormality*, *location*, *severity*, *diagnosis*, *reasoning*, and *treatment*. For normal cases, only one QA pair related to abnormality detection was generated. The complete prompt templates are provided in (Appendix A (Figures 9–10).)^{Deepali-edit}

A textual similarity analysis between train and test questions (Figure 4) confirmed adequate lexical diversity, with a mean TF-IDF cosine similarity of 0.573 and a mean Jaccard similarity of 0.268, indicating limited word overlap and ensuring generalization. Qualitative examples of train and test question-answer pairs are shown in (Appendix C (Table 13 and Table 14))^{Deepali-edit}.

4.2 Stage 2: Expert Validation and Data Cleaning

Subsequent to image selection and automated QA generation, the dataset was clinically verified by a panel of **10 practicing doctors**, comprising five clinicians from **India** and five from **Thailand**. All participating clinicians were within an age range of **32–55 years**, fluent in English, and experienced in routine interpretation

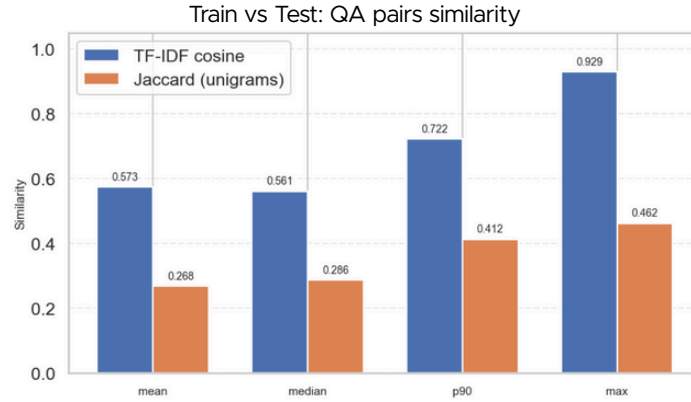


Fig. 4. Train–test lexical similarity analysis of **SpineXR-VQA** question–answer pairs. TF–IDF cosine and Jaccard unigram similarity were computed between each test question–answer and its closest training counterpart. The 90th percentile (p90) denotes the upper similarity bound for 90% of test questions, while *max* represents the most similar question pair. The results (mean cosine 0.57, Jaccard 0.27; p90 cosine 0.72) confirm that the test set contains rephrased but non-duplicate questions, ensuring generalization.

of spine X-rays. Each clinician received an honorarium of **INR 4,000** (India) or **THB 4,000** (Thailand) for their participation. The verification was conducted through a structured questionnaire implemented using Google Forms, where clinicians reviewed each spine X-ray image together with its associated question–answer pairs. In the standard setting, each image–QA pair was reviewed by a single clinician. To assess inter-annotator reliability, a subset of **283 images** corresponding to **994 QA pairs** was independently reviewed by two clinicians. In cases of disagreement between the two reviewers, a third senior clinician was consulted to adjudicate, and the incorrect or lower-confidence QA pair was replaced with the corrected, expert-validated version. A representative sample of Google Form for the clinician validation is provided in the **(Appendix D (Figure 11))**^{Deepali–edit}.

Each form presented one X-ray image with corresponding QA pairs, divided into dedicated sections per image. For every question and answer, experts selected one of three options: *Question Correct*, *Question Not Correct*, or *Not Sure* for the questions, and similarly *Answer Correct*, *Answer Not Correct*, or *Not Sure* for the answers. When either the question or answer was marked as “Not Correct,” the medical expert was required to provide a corrected version in a short-answer text field.

During data cleaning, QA pairs labeled as “Not Sure” (‘Question not sure’: 74, ‘Answer not sure’: 234) without an accompanying correction were dropped. For entries where the reviewers provided corrected questions or answers, those were incorporated into the final dataset. This validation ensured that only fully reviewed, expert-confirmed QA pairs were retained. **(Upon acceptance, the cleaned and verified dataset will be publicly released via Hugging Face. The dataset structure and fields are mentioned in Appendix B (Table 12))**^{Deepali–edit} **(R3_comment3-I am not able to understand what sample should I put the screen shot of the csv file)**^{Deepali–Comment} **(One example for training set and test set is shown in Figure5)**^{Deepali–edit} along with the image.**(R3-comment5)**^{Deepali–Comment}

Example 1: Image ID – vindr_train_015.png			Example 2: Image ID – vindr_test_538.png		
					
Category	Question	Answer	Category	Question	Answer
Abnormality	What is the nature of the abnormality observed in the X-ray image?	The X-ray image shows disc space narrowing at the C5–C6 and C6–C7 levels, indicating degenerative changes in the intervertebral discs.	Abnormality	Can the X-ray image be evaluated for any radiographic abnormalities?	The X-ray image shows multiple abnormalities, including vertebral collapse, osteophyte formation, and the presence of a surgical implant.
Severity	What is the severity of the disc space narrowing observed in the X-ray image?	The disc space narrowing appears moderate, with a noticeable reduction in the height of the intervertebral discs at the affected levels.	Severity	Assess the degree of vertebral compression.	The vertebral collapse appears moderate to severe, with significant loss of vertebral height and potential compromise of spinal stability.
Location	Where is the disc space narrowing observed in the X-ray image?	The disc space narrowing is observed at the C5–C6 and C6–C7 levels, located in the cervical spine.	Location	Can you identify the specific areas within the image that show abnormalities?	The abnormalities are located in the lumbar spine, primarily at the L1–L2 and L2–L3 levels.
Diagnostic	What is the most likely diagnosis based on the X-ray image?	The most likely diagnosis is degenerative disc disease, characterized by progressive deterioration of the intervertebral discs.	Diagnostic	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is degenerative disc disease with vertebral collapse and osteophyte formation, possibly complicated by prior surgical intervention.
Reasoning	What are the possible causes of the disc space narrowing observed in the X-ray image?	Possible causes include age-related degeneration, trauma, or underlying conditions such as osteoarthritis or spinal stenosis.	Reasoning	Identify the underlying factors contributing to the observed abnormalities.	Potential contributing factors include degenerative disc disease, osteoporosis, prior trauma, or infection.
Treatment	What are the suggested treatment or management options for the detected abnormalities?	Suggested management includes conservative treatment with physical therapy, pain medication, and lifestyle modifications, with surgical intervention considered in severe cases.	Treatment	Given the presence of detected abnormalities, what treatment or management strategies are typically recommended?	Recommended management includes conservative treatment with analgesics and physical therapy, surgical stabilization if indicated, and regular follow-up to monitor disease progression.

Fig. 5. Sample examples of Train set and Test set

5 SpineXR-VQA Analysis

5.1 Statistics and Coverage

The final cleaned version of **SpineXR-VQA** comprises 8,272 validated QA pairs spanning 2,187 unique spine X-rays (Table 3). The dataset maintains a balanced distribution across both countries and splits, with no image overlap between train and test subsets. A total of 283 images are intentionally shared between the Indian and Thai training sets to facilitate cross-institutional consistency analysis, yielding 994 aligned QA pairs for comparative evaluation. Such alignment enables country-wise reasoning comparison without compromising the integrity of test data separation.

Table 3. Final cleaned distribution of **SpineXR-VQA** across countries and splits. No image overlaps exist between Train and Test. A total of 283 images are shared between *India-Train* and *Thailand-Train* for cross-country comparison.

Country	Split	Unique Images	QA Pairs
India	Train	879	3,266
India	Test	150	614
Thailand	Train	1,019	3,951
Thailand	Test	139	441
Total (cleaned)	–	2,187	8,272
<i>Common images (India-Train $\cap$ Thailand-Train)</i>		283	994

Table 4. (Distribution of QA pairs across categories and dataset splits with totals)

Category	India Test	India Train	Thailand Test	Thailand Train	Total
Abnormality	150	878	129	1029	2,186
Diagnostic	93	477	64	578	1,212
Location	92	477	59	573	1,201
Reasoning	93	478	68	600	1,239
Severity	93	478	52	566	1,189
Treatment	93	478	69	605	1,245
Total	614	3,266	441	3,951	8,272

(Figure 6 presents the image-level distribution of spinal abnormalities in the SpineXR-VQA dataset, computed by aggregating unique abnormalities per image using rule-based extraction from answer text. Normal findings dominate (33.7%), followed by fractures (21.0%). Less frequent conditions grouped as *Other* (<10%) account for 18.7%, while degeneration (14.3%) and osteophytes (12.3%) constitute notable proportions. The *Other* category includes alignment abnormalities, disc space narrowing, foraminal stenosis, spinal deformities, spondylolisthesis, implant/hardware cases, and vertebral collapse.)

(As shown in Table 5, these abnormalities span diverse clinical question categories, with fractures and degenerative conditions prominently contributing to diagnostic, reasoning, and severity-related queries. Overall, the distribution reflects a realistic clinical spectrum, with a balanced representation of normal and pathological findings.)

(Table 4 summarizes the distribution of QA pairs across categories and dataset splits. The dataset comprises 8,272 QA pairs, with a larger proportion allocated to the training sets. While the Abnormality category has the highest representation, the remaining categories are relatively balanced across splits, reflecting the structured annotation design of the dataset.)

Table 5. Distribution of abnormalities across question categories (image-level).

Abnormality	Diagnostic	Location	Reasoning	Severity	Treatment
Degeneration	392	306	362	345	119
Fracture	569	529	571	540	298
No finding / Normal	120	101	118	112	55
Osteophytes	341	284	283	316	96
Other (<10%)	504	415	474	466	211

Table 6 provides basic statistics of **SpineXR-VQA**. Question lengths range from 5–29 words, remaining concise and task-focused, while answer lengths extend up to 129 words, reflecting the need for descriptive clinical explanations rather than short factual responses. Notably, test-set questions exhibit greater lexical variation than training and validation sets, consistent with their generation using reasoning-oriented prompts.

While word-length statistics alone do not capture linguistic diversity, they indicate a clear distinction between concise query formulation and elaborative answer construction. Together with full clinician verification and

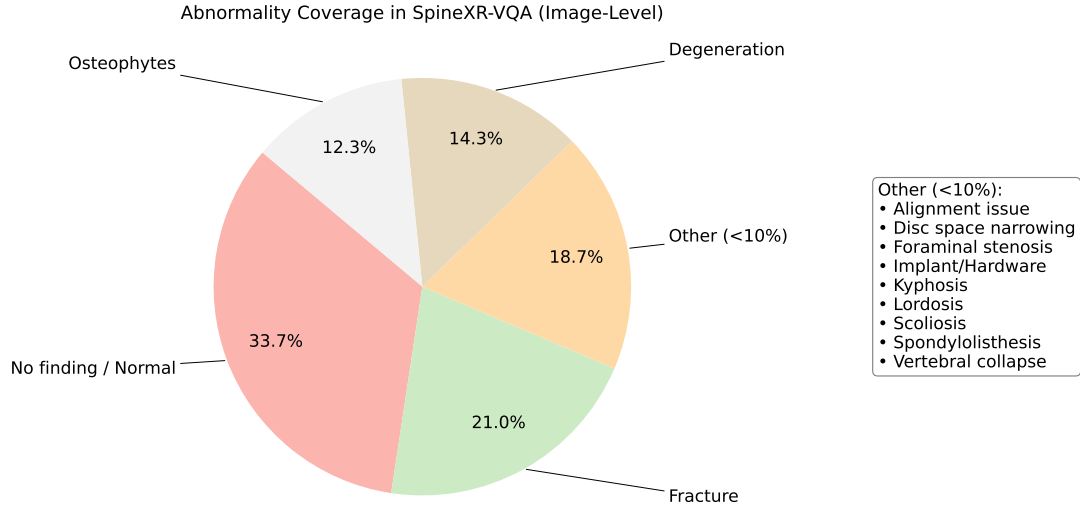


Fig. 6. Image-level distribution of spinal abnormalities in the **SpineXR-VQA** dataset, computed by aggregating unique abnormalities per image from answer text. Normal findings constitute the largest proportion (33.7%), followed by fractures (21.0%). Degeneration (14.3%) and osteophytes (12.3%) also contribute substantially. Less frequent abnormalities are grouped under *Other (<10%)* (18.7%), including alignment abnormalities, disc space narrowing, foraminal stenosis, spinal deformities, spondylolisthesis, implant/hardware cases, and vertebral collapse. The distribution reflects a balanced mix of normal and pathological findings, supporting clinical diversity.

Table 6. Question and answer length statistics across Train and Test splits of **SpineXR-VQA**. Values represent word-level minima and maxima aggregated across India and Thailand subsets.

Split	Q_min_words	Q_max_words	A_min_words	A_max_words
Train	5	17	3	129
Test	6	29	3	115

coverage of multiple diagnostic reasoning categories, these characteristics support the use of **SpineXR-VQA** as a benchmark for evaluating open-ended, clinically grounded reasoning in spinal radiography.

5.1.1 Cross-Country Agreement on Shared Images (283 radiographs, 994 aligned QA pairs). To quantify cross-institution consistency, we aligned **994** QA pairs across the **283** images that appear in both India-Train and Thailand-Train. Table 7 summarizes agreements for *questions* and *answers*. High *question* agreement on correctness (86.0%) indicates that both clinical teams judged the prompts and question framing similarly, with minimal cross-country disagreement (mismatch: 0.9%). In contrast, *answers* show substantially lower agreement on correctness (65.9%) and a much larger mass of *agree (wrong)* cases (31.4%), indicating that both teams often concurred that the model-generated answers were incorrect. Residual answer *mismatch* (2.7%) is small but non-zero, reflecting cross-institution nuance in phrasing or emphasis. Overall, these results validate the dataset’s cross-country reliability while underscoring that VLM-generated *answers* require thorough expert verification.

Table 7. Cross-country agreement on the 994 aligned QA pairs from the 283 shared images. Values are counts (percent).

Relation	Questions	Answers
Agree (correct)	855 (86.0%)	655 (65.9%)
Agree (wrong)	130 (13.1%)	312 (31.4%)
Mismatch	9 (0.9%)	27 (2.7%)
Total	994	994

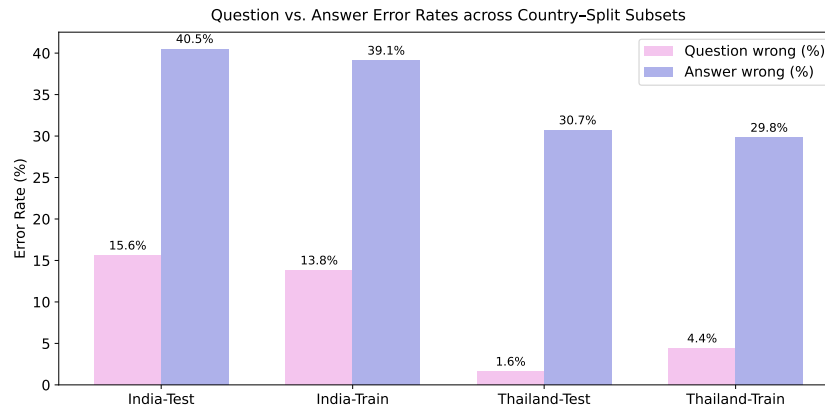


Fig. 7. Question and answer error rates across country-split subsets for QA pairs generated by **LLaMA 3.2 Vision-Instruct**. Answer disagreement rates (30–41%) consistently exceed question disagreement rates (2–16%), particularly in Indian subsets, highlighting the model’s tendency to produce syntactically coherent but semantically imprecise responses.

5.2 Error Analysis of QA Pairs Generated by LLaMA 3.2 Vision-Instruct

The distribution of question and answer error rates across country-split subsets (Figure 7) provides a quantitative view of the behavior of the vision-language model **LLaMA 3.2 Vision-Instruct** during QA generation. Here, an *error* (or *disagreement*) is defined as a question or answer that was marked as *Not Correct* by clinician during validation process. Across all subsets, answer error rates (30–41%) are consistently two to three times higher than question error rates (2–16%), indicating that the model is substantially more reliable at generating syntactically and semantically appropriate questions than at producing clinically correct answers. This discrepancy arises, possibly because question generation is largely a structural task guided by the prompt template, whereas answer generation requires domain-grounded reasoning. The high proportion of incorrect answers—particularly in the Indian splits (A_{wrong} : 39–41%)—shows that even when the model recognizes relevant visual cues, it often fails to contextualize them into accurate diagnostic statements. In several cases, the model defaulted to normal findings or vague terms such as “mild degeneration,” reflecting over-generalization and a lack of lesion-specific understanding.

The comparatively lower Thai error rates (A_{wrong} : 29–31%) suggest that responses were more consistent in phrasing and template alignment, though still limited by shallow semantic reasoning. Overall, these results confirm that LLaMA 3.2 is capable of capturing visual and lexical patterns but lacks the medical expertise needed for diagnostic inference. A closer examination of the dataset reveals that the model performs acceptably at *recognition-level grounding*—identifying abnormalities in an image—but fails at *reasoning-level interpretation*,

where clinical validation and anatomical precision are required. Given that over one-third of the generated answers required manual correction, these findings demonstrate that large vision–language models cannot yet be relied upon for autonomous medical QA pair generation without expert oversight. Some of the sample examples where the model failed or hallucinated is shown in Table 8. It also highlights the necessity of clinician-verified datasets such as **SpineXR-VQA**, which quantify and constrain the reliability gap between automated generation and expert validation.

5.2.1 Lexical and Linguistic Complexity of Questions and Answers. To ensure that the **SpineXR-VQA** dataset is linguistically coherent yet sufficiently challenging for multimodal reasoning, we analyze its lexical and structural properties across the **Train/Test** and **India/Thailand** splits. Although question–answer pairs were generated by a single vision–language model, the India/Thailand stratification reflects independent expert validation subsets and allows us to verify that linguistic characteristics remain stable across geographically distinct clinical validation processes. Specifically, we evaluate mean length, vocabulary richness, lexical density, and readability. The results are summarized in Figure 8, which visualizes token lengths, vocabulary diversity (Type–Token Ratio and MTLD), and lexical density comparisons across splits and countries.

Lexical Complexity Metrics. We characterize the linguistic properties of **SpineXR-VQA** using four complementary lexical complexity measures, each capturing a different aspect of text structure and difficulty (Figure 8).

- (1) **Mean length.** Mean token length provides a coarse measure of descriptive depth and controls for trivial cues that may arise from overly short or excessively verbose responses. Let L_d denote the token length of an instance d . The mean lengths for questions and answers are defined as:

$$\bar{L}_Q = \frac{1}{|Q|} \sum_{q \in Q} L_q, \quad \bar{L}_A = \frac{1}{|A|} \sum_{a \in A} L_a.$$

As shown in Figure 8(a), question lengths remain tightly clustered (approximately 10–13 tokens), while answers are consistently longer (25–30 tokens) across all splits. This controlled separation reflects the intended design: concise prompts paired with explanatory, clinically meaningful answers.

- (2) **Type–Token Ratio (TTR).** Type–Token Ratio captures surface-level lexical variety by measuring the proportion of unique words in a corpus. For a token multiset T and unique vocabulary set U , TTR is defined as:

$$\text{TTR} = \frac{|U|}{|T|}.$$

While TTR values are slightly higher for the test split (Figure 8(b)), this increase is expected due to the intentional lexical diversification introduced by Prompt B. However, because TTR is sensitive to corpus size, it is interpreted in conjunction with MTLD.

- (3) **Measure of Textual Lexical Diversity (MTLD).** MTLD provides a length-robust estimate of sustained lexical diversity by computing how many tokens are processed, on average, before the running TTR falls below a threshold τ (set to 0.72). If N tokens yield F such factors, MTLD is given by:

$$\text{MTLD} = \frac{N}{F}.$$

Comparable MTLD values across geographic subsets and a moderate increase in the test split indicate that lexical diversity is consistently maintained, with the test set presenting a slightly richer vocabulary without introducing distributional shifts.

- (4) **Lexical density.** Lexical density measures the proportion of content-bearing words, offering insight into informational load. Let C_d denote the number of content words (nouns, verbs, adjectives, and adverbs) in

instance d . For a subset S , lexical density is computed as:

$$LD(S) = \frac{\sum_{d \in S} C_d}{\sum_{d \in S} L_d}.$$

Figures 8(c–d) show highly consistent lexical density values (approximately 0.55–0.60) across both country and train–test splits, suggesting that the informational content of questions and answers remains stable and is not confounded by stylistic variation.

Collectively, these analyses demonstrate that **SpineXR-VQA** maintains controlled question and answer lengths, stable lexical density, and consistent vocabulary richness across geographic and train–test splits. At the same time, the test set exhibits mildly increased lexical and readability complexity, confirming that it poses a more challenging linguistic setting without introducing distributional artifacts. These properties support the use of **SpineXR-VQA** as a reliable benchmark for evaluating generalization and clinically grounded reasoning in medical visual question answering.

Table 8. Qualitative error analysis of LLaMA-generated QA pairs across representative cases.

QA Type	Question (✓/✗)	Answer (✓/✗)	Corrected Answer
Example A (Image: vindr_train_095.png): Case Where the Model Was Fully Successful			
Abnormality	What is the nature of the abnormality observed in the X-ray image? ✓	The X-ray image shows a fracture in the lumbar spine, specifically at the L4–L5 vertebrae. The fracture appears to be a compression fracture, characterized by a loss of height in the vertebral body. ✓	–
Severity	What is the level of severity diagnosed? ✓	The fracture is classified as a compression fracture, which is typically considered a stable fracture. However, the severity can vary depending on the degree of compression and the presence of any associated injuries. ✓	–
Location	Where is the fracture located in the X-ray image? ✓	The fracture is located in the lumbar spine, specifically at the L4–L5 vertebrae. This region is prone to fractures due to its weight-bearing nature and the stress it undergoes during daily activities. ✓	–
Diagnostic	What is the most likely diagnosis based on the X-ray image? ✓	The most likely diagnosis is a compression fracture of the L4–L5 vertebrae, which can be caused by various factors such as osteoporosis, trauma, or degenerative disc disease. ✓	–
Reasoning	What are the possible causes of the fracture observed in the X-ray image? ✓	The possible causes of the fracture include osteoporosis, trauma, or degenerative disc disease. Osteoporosis can weaken the bones, making them more susceptible to fractures. Trauma, such as a fall or a car accident, can cause a sudden impact that leads to a fracture. Degenerative disc disease can cause the discs to break down, leading to a compression fracture. ✓	–

QA Type	Question (✓/✗)	Answer (✓/✗)	Corrected Answer
Treatment	What is the suggested treatment or management option for the detected fracture? ✓	The suggested treatment or management option for the detected fracture is conservative management, which includes rest, pain management, and physical therapy. In some cases, surgery may be necessary to stabilize the spine and alleviate pain. ✓	–
Example B (Image: vindr_train_189.png): Case Where the Model Was Incorrect			
Abnormality	What is the nature of the abnormality observed in the X-ray image? ✓	The X-ray image reveals a fracture in the lumbar spine, specifically affecting the L3 vertebra. This is evident from the irregularity in the bone structure and the presence of a clear breakage line. ✗	No fracture. Decreased disc height at L4/5.
Severity	What is the severity of the fracture observed in the X-ray image? ✓	The fracture appears to be a significant one, with a substantial displacement of the bone fragments. This suggests that the injury may have resulted in a substantial loss of spinal stability and potentially caused nerve damage or compression. ✗	No fracture seen. Decreased disc height at L4/5.
Location	Where is the fracture located in the X-ray image? ✓	The fracture is located in the lumbar spine, specifically affecting the L3 vertebra. This region is crucial for supporting the body's weight and facilitating movement, making the fracture potentially debilitating. ✗	No fracture. Disc height loss at L4/5.
Diagnostic	What is the most likely diagnosis based on the X-ray image? ✓	The most likely diagnosis is a lumbar spine fracture, specifically affecting the L3 vertebra. This diagnosis is supported by the clear visualization of the fracture line and the displacement of the bone fragments. ✗	Degenerative disc disease.
Reasoning	What are the possible causes of the fracture observed in the X-ray image? ✓	The possible causes of the fracture include a traumatic injury, such as a fall or a car accident, which may have resulted in a sudden and forceful impact on the lumbar spine. Other possible causes include osteoporosis and degenerative process. ✓	–
Treatment	What is the suggested treatment or management option for the detected fracture? ✓	The suggested treatment or management option for the detected fracture is immediate medical attention, including immobilization of the affected area and pain management. In some cases, surgery may be necessary to stabilize the spine and prevent further complications. ✓	–
Example C (Image: vindr_train_199.png): Case Where the Model Hallucinated a Fracture			

QA Type	Question (✓/✗)	Answer (✓/✗)	Corrected Answer
Abnormality	Is there any abnormality present in this image? ✓	Yes, there is an abnormality present in this image. The X-ray image shows a fracture of the L1 vertebra, characterized by a clear break in the vertebral body and the presence of a callus, indicating a healing process. The fracture appears to be a compression fracture, which is a common type of spinal fracture that occurs when the bone is compressed or crushed. ✗	No abnormality seen.
Severity	What is the severity of the fracture? ✓	The severity of the fracture is moderate. The fracture is stable, and there is no evidence of significant displacement or rotation of the vertebral body. However, the fracture is still considered unstable due to the presence of a callus, which indicates that the bone is still in the process of healing. ✗	No abnormality seen.
Location	Where is the fracture located in the image? ✓	The fracture is located in the L1 vertebra, which is the first lumbar vertebra in the lower back. The fracture is visible in the midline of the spine, and it appears to be affecting the anterior aspect of the vertebral body. ✗	No abnormality seen.
Diagnostic	What is the most likely diagnosis for the fracture? ✓	The most likely diagnosis for the fracture is a compression fracture of the L1 vertebra. This type of fracture is commonly caused by osteoporosis, which is a condition characterized by weak and brittle bones. The fracture may also be caused by trauma, such as a fall or a car accident. ✗	No abnormality seen.
Reasoning	What are the possible causes of the fracture? ✓	The possible causes of the fracture include osteoporosis, trauma, and other underlying medical conditions that may have weakened the bone. Osteoporosis is a common cause of compression fractures, as it can lead to a loss of bone density and strength. Trauma, such as a fall or a car accident, can also cause a compression fracture by applying excessive force to the spine. ✗	No abnormality seen.
Treatment	What is the suggested treatment or management option for the fracture? ✓	The suggested treatment or management option for the fracture is conservative management, which includes rest, pain management, and physical therapy. In some cases, surgery may be necessary to stabilize the spine and prevent further injury. The patient may also be prescribed medication to help manage pain and prevent further fractures. ✗	No abnormality seen.

6 Model Benchmarking

Given the recent progress in multimodal machine learning and medical imaging, we evaluate the performance of MLLMs on SpineXR-VQA. The VQA task is as follows: given an X-ray image and a corresponding question

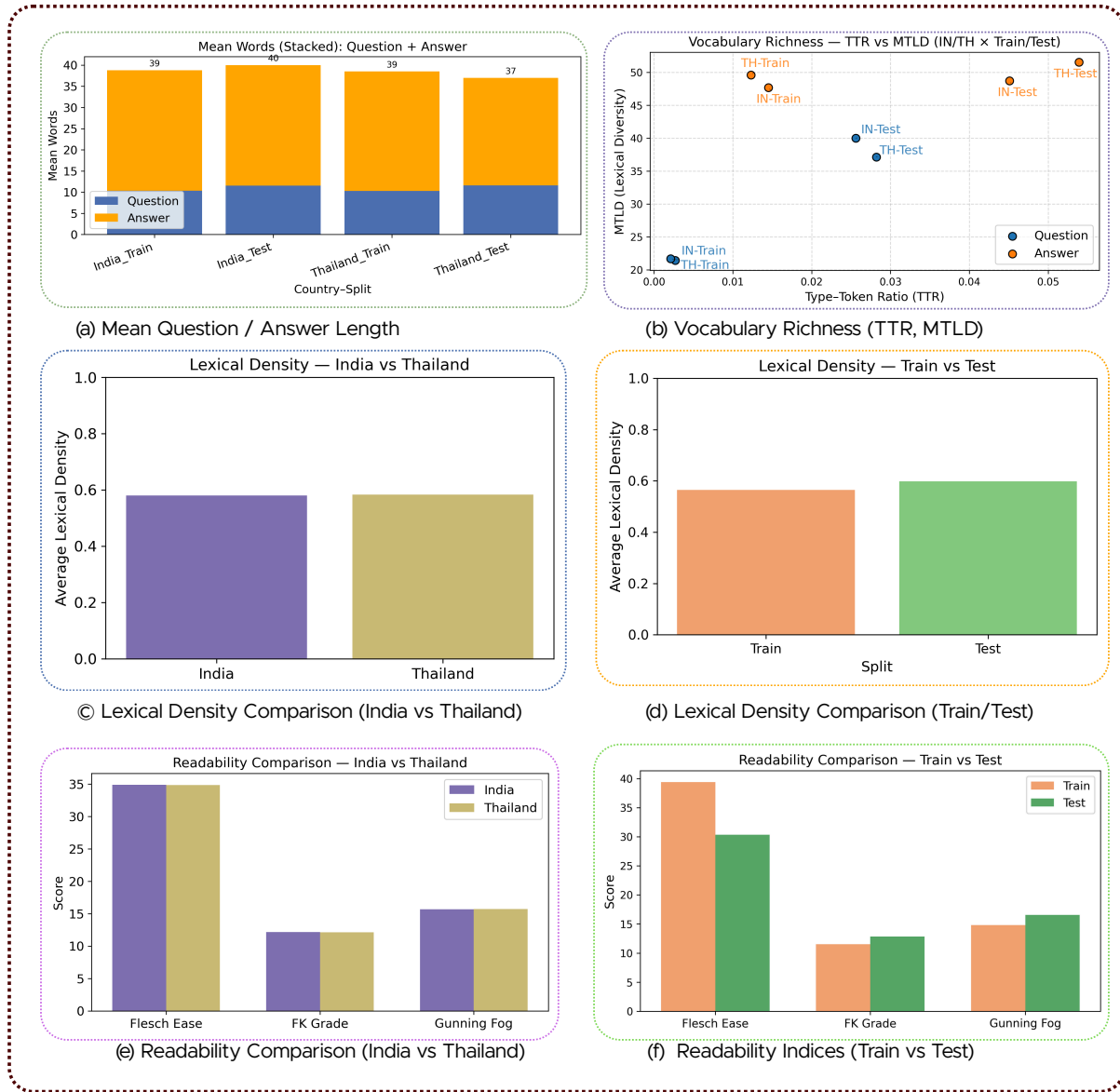


Fig. 8. Lexical characteristics of **SpineXR-VQA**. (a) Mean question/answer length (tokens); (b) Vocabulary richness via TTR and MTLD; (c) Lexical density by country (India vs. Thailand); (d) Lexical density by split (Train vs. Test). Together, these panels show controlled length, stable richness across subsets, and similar content-word density—supporting linguistic balance without leakage.

related to the image, a model is required to generate a descriptive answer for the question. Before describing the experiments and results, we first describe the evaluation metrics used for benchmarking.

Table 9. Benchmarking results on **SpineXR-VQA** across proprietary, general-purpose, and medical vision-language models. Columns are ordered from most to least semantically important. Within each group, the best value per metric is shaded and the best model (by BERTScore-F1) is underlined.

Model name	BERTScore-F1	Semantic Similarity (BioClinicalBERT)	ROUGE-L	METEOR	BLEU-4	TF-ID
Zero Shot Experiments						
Proprietary Models						
Claude Sonnet [3]	0.5876	0.9090	0.2038	0.1492	0.0469	0.0808
GPT-4.1 [32]	0.5670	0.9071	0.1737	0.1221	0.0167	0.0678
O4 Mini [32]	0.5467	0.7839	0.1443	0.0711	0.0064	0.0460
Claude Haiku[3]	0.5459	0.8926	0.1645	0.1185	0.0249	0.0626
GPT-5.1 [32]	0.5323	0.8955	0.1256	0.0820	0.0065	0.0571
GPT-5 Mini [32]	0.5126	0.8368	0.1094	0.0599	0.0017	0.0454
Gemini Flash [9]	0.4761	0.9022	0.1438	0.1478	0.0121	0.0644
General Purpose Open Models						
DeepSeek [42]	0.3705	0.9178	0.2061	0.1739	0.0317	0.0681
LLaVA-Phi[50]	0.3075	0.9157	0.1629	0.2077	0.0239	0.0806
Qwen 2.5 [35]	0.2917	0.9076	0.1584	0.2161	0.0217	0.0890
BLIP-2[22]	0.0860	0.8193	0.1231	0.0585	0.0016	0.0258
Medical Models						
Lingshu[40]	0.5303	0.8672	0.1257	0.1586	0.0148	0.0432
MedGemma [38]	0.5202	0.8782	0.1624	0.0952	0.0153	0.0564
MedFlamingo [30]	0.4129	0.7946	0.0750	0.0974	0.0079	0.0426
LLaVA Med [21]	0.3626	0.9209	0.2152	0.2044	0.0424	0.0820
Fine-tuned Models Experiments						
General Purpose Open Models						
BLIP-2[22]	0.5472	0.9432	0.4109	0.3465	0.1809	0.1803
Qwen 2.5 [35]	0.5417	0.9414	0.4303	0.3943	0.1815	0.2413
LLaVA-Phi[50]	0.3709	0.9084	0.2277	0.2860	0.0733	0.1448
Medical Models						
MedGemma[38]	0.5481	0.9444	0.4349	0.0974	0.1082	0.2830
Lingshu[40]	0.5463	0.8981	0.1562	0.1628	0.0221	0.0373
MedFlamingo [30]	0.5144	0.8882	0.1506	0.1969	0.0351	0.0933
LLaVA Med[21]	0.3012	0.8778	0.1424	0.2534	0.0457	0.1584

6.1 Evaluation Metrics

While there is no one single metric that will capture model performance reliably, to comprehensively evaluate model performance on **SpineXR-VQA**, based on current research, we employ five complementary metrics that capture lexical overlap, structural similarity, and semantic fidelity. Note even these metrics are not perfect and coming up with a reliable metric(s) is a research direction on its own and we plan to explore this in the future.

Metrics are ordered from most to least semantically informative to reflect the generative and reasoning-oriented nature of the task.

BERTScore-F1: BERTScore [47] evaluates semantic similarity between a generated answer and a reference answer by comparing contextual token embeddings obtained from a pretrained language model. Let $\{\mathbf{h}_i\}_{i=1}^n$ and $\{\hat{\mathbf{h}}_j\}_{j=1}^m$ denote the embeddings of the generated and reference tokens, respectively. Token-level similarity is computed using cosine similarity:

$$\text{sim}(i, j) = \frac{\mathbf{h}_i \cdot \hat{\mathbf{h}}_j}{\|\mathbf{h}_i\| \|\hat{\mathbf{h}}_j\|}.$$

BERTScore is defined using token-level maximum similarity matching between the two sequences, from which precision and recall are computed as:

$$P = \frac{1}{n} \sum_{i=1}^n \max_{1 \leq j \leq m} \text{sim}(i, j), \quad R = \frac{1}{m} \sum_{j=1}^m \max_{1 \leq i \leq n} \text{sim}(i, j).$$

The BERTScore-F1 is then computed as the harmonic mean of precision and recall:

$$\text{BERTScore-F1} = \frac{2PR}{P + R}.$$

By leveraging contextual embeddings and intrinsic token-level matching, BERTScore-F1 is robust to paraphrasing and lexical variation, making it suitable for evaluating long-form clinical text generation.

Semantic Similarity (BioClinicalBERT)[2] : To assess domain-specific semantic alignment, we compute the cosine similarity between BioClinicalBERT sentence embeddings [2] of the generated and reference answers:

$$\text{SemSim} = \cos(\mathbf{e}_{\text{pred}}, \mathbf{e}_{\text{ref}}).$$

Because BioClinicalBERT is trained on large-scale clinical corpora, this metric is sensitive to medically meaningful phrasing and is particularly suitable for evaluating radiology-style explanations.

ROUGE-L: ROUGE-L [23] measures the longest common subsequence (LCS) between prediction and reference, capturing structural and narrative similarity independent of exact n-gram matching. It is defined as:

$$\text{ROUGE-L} = \frac{\text{LCS}(X, Y)}{\max(|X|, |Y|)}.$$

This metric is effective for multi-sentence clinical descriptions and summary-style answers.

METEOR: METEOR [5] computes a weighted harmonic mean of unigram precision and recall while incorporating stemming, synonym matching, and paraphrase alignment. The metric is formulated as:

$$\text{METEOR} = F_\alpha \cdot (1 - \text{penalty}),$$

where F_α denotes the tunable F-score. METEOR captures partial matches and semantic variants more effectively than BLEU, making it suitable for evaluating free-text radiology explanations.

BLEU-4 [33]: BLEU-4 measures modified n-gram precision up to four-grams using the geometric mean combined with a brevity penalty:

$$\text{BLEU-4} = \text{BP} \cdot \exp\left(\frac{1}{4} \sum_{n=1}^4 \log p_n\right),$$

where p_n denotes modified n-gram precision. While standard in machine translation, BLEU is less sensitive to semantic equivalence in generative clinical VQA; thus, we treat it as a complementary indicator of surface-level lexical overlap.

6.2 Performance of MLLMs

While several multimodal large language models (MLLMs) and medical vision–language models (VLMs) have been proposed in recent years [10, 11, 24, 37, 44, 46], due to limited compute budgets and deployment costs, we evaluate only a subset of prominent models that have been widely reported in prior work. These models are representative of contemporary state-of-the-art systems used in medical and general-domain vision–language reasoning. Despite strong benchmark performance reported for recent multimodal LLMs [21, 31, 50], our experiments reveal substantial and systematic failure modes across all model categories on **SpineXR-VQA** (**Table 9**). **Two sample examples for each model output is presented in Appendix F (Table 16))***Deepali–edit*.

State-of-the-art proprietary systems, such as Claude Sonnet⁴ and GPT-4.1⁵, frequently produce answers that resemble plausible radiology reports but are weakly grounded in the visual evidence provided by the input X-ray. As shown in Table 9, these models achieve moderately high BERTScore-F1 values (0.53–0.59) and high embedding-based semantic similarity scores computed using BioClinicalBERT (often exceeding 0.9). However, their lexical overlap metrics—BLEU-4, ROUGE-L, and TF-IDF cosine similarity—remain consistently low. This divergence suggests that high embedding-level semantic similarity does not necessarily imply faithful use of clinically precise terminology or accurate alignment with expert-validated reference answers. **(We further perform a category-wise evaluation across clinically relevant tasks (*Abnormality, Severity, Location, Diagnostic, Reasoning, Treatment*) for all models using multiple metrics. Detailed results are presented in Appendix G (Figures 12–16). Overall, consistent trends are observed across metrics: performance is relatively strong for *Abnormality, Severity, and Location*, but degrades notably for *Reasoning and Treatment*, highlighting challenges in clinically grounded inference. General-purpose and medical models: particularly fine-tuned variants such as Qwen 2.5 (FT), BLIP-2 (FT), and MedGemma (FT) consistently rank among the top performers. In contrast, proprietary models often achieve high semantic similarity while underperforming on lexical metrics, indicating weaker alignment with clinically precise terminology.)***Deepali–edit* **(R1_Comment2)***Deepali–Comment* Importantly, we do not interpret semantic similarity as a measure of surface-level linguistic overlap. Instead, semantic similarity captures embedding-space proximity, which may arise from shared radiology templates and common diagnostic phrasing, even when clinically critical details are incorrect or missing. In contrast, lexical metrics such as BLEU, ROUGE, and TF-IDF are more sensitive to deviations in medically salient terms (e.g., vertebral level, fracture morphology, or severity descriptors). As observed in Table 9, proprietary models often exhibit high semantic similarity alongside low TF-IDF scores, indicating that semantically adjacent but clinically incorrect phrasing is frequently rewarded by embedding-based metrics. These discrepancies manifest in failure modes such as hallucinated abnormalities, incorrect anatomical localization, and substitution of degenerative findings for acute pathology, as illustrated in **(Table 17 in Appendix H)***Deepali–edit* **(R2-comment_1)***Deepali–Comment* and discussed in the model error analysis section.

General-purpose open vision–language models perform considerably worse. Zero-shot evaluation of models such as DeepSeek-VL [42], LLaVA-Phi [50], and Qwen 2.5 [35] reveals a strong tendency to default to generic radiology statements (e.g., “degenerative changes noted,” “no obvious abnormality”) and an inability to extract fine-grained spinal features such as osteophytes, disc space narrowing, vertebral alignment, or fracture morphology. These models consistently obtain low scores across lexical metrics (low TF-IDF similarity), reflecting weak domain grounding. While fine-tuning improves performance for models such as BLIP-2 (FT) [22] and Qwen 2.5 (FT), their outputs remain verbose yet clinically imprecise, suggesting that conventional VLM architectures struggle to internalize subtle radiographic cues.

⁴<https://claude.ai/new>

⁵<https://chatgpt.com/>

Surprisingly, medical VLMs evaluated in the zero-shot setting—MedFlamingo [30], LLaVA-Med [21], MedGemma [38], and Lingshu [40]—do not consistently outperform general-purpose models. Despite domain-specific pretraining, these models often exhibit overgeneralization, producing high-level diagnostic phrases (e.g., “degenerative changes” or “normal alignment”) irrespective of image-specific pathology. After supervised fine-tuning, medical models demonstrate improved alignment with expert reasoning; however, even the strongest model, MedGemma (FT), which achieves the highest semantic similarity score (0.9444), continues to exhibit clinically relevant errors such as misclassification of severity or conflation of spinal regions. The relatively modest TF-IDF and BLEU scores for these models further underscore the limitations of relying on semantic metrics in isolation.

Collectively, these findings reveal critical shortcomings across model classes: (1) inadequate image grounding and anatomical localization, (2) failure to detect subtle degenerative or traumatic features, (3) overreliance on memorized radiology templates, and (4) limited clinical reasoning under domain shift. While fine-tuning substantially improves performance across both semantic and lexical metrics, persistent gaps highlight the need for spine-specific supervision signals, structured evaluation protocols, and domain-aligned multimodal representations. Detailed qualitative examples across question categories and models are presented in the model error analysis section. **(Table 17 provides error cases explicitly identified and annotated by manual expert reviewers. Together, these tables further substantiate the observed failure patterns and qualitative trends.)***Deepali-edit*

6.3 Model Output Error Analysis

For qualitative error analysis, we selected five representative models spanning distinct modeling paradigms, each chosen based on the highest semantic similarity scores within its respective category: (i) a proprietary model evaluated in a zero-shot setting, (ii) a general-purpose model in a zero-shot setting, (iii) a medical-domain model in a zero-shot setting, (iv) a general-purpose fine-tuned model, and (v) a medical-domain fine-tuned model.

(Using the common test split shared across all models, we manually evaluated outputs corresponding to 35 spine X-ray images (10 normal and 25 abnormal), yielding a total of 160 question–answer pairs. The evaluation was complemented by a fine-grained annotation of clinically relevant error types observed in model outputs. As summarized in Appendix H (Table 17), recurring error patterns include false-negative detection (missed abnormalities), pathology substitution (confusing acute findings with degenerative conditions), anatomical mislocalization (incorrect spinal level or region), reasoning collapse (failure to correctly interpret visual evidence), and unsafe management recommendations. This structured analysis provides a quantitative characterization of clinically meaningful failure modes that are not captured by standard lexical or embedding-based metrics.)*Deepali-edit* **(R2-commnet2 and R3_comment_2)***Deepali-Comment*

Despite their strong semantic similarity performance, all evaluated models exhibit clinically significant failure modes when compared against expert-validated ground truth **((Table 17 in Appendix H))***Deepali-edit* **(R2-Comment_1)***Deepali-Comment*. These errors are observed as pathology substitution (e.g., predicting degenerative disc disease in place of acute fractures), anatomical mislocalization across spinal regions, and false-negative findings despite radiographically evident pathology. In addition, several models produce unsafe or incomplete management recommendations for unstable injuries, revealing a critical disconnect between fluent language generation and clinically sound reasoning. Collectively, these results demonstrate that semantic similarity alone is insufficient to ensure diagnostic accuracy or clinical reliability, underscoring the need for clinician-centered evaluation beyond surface-level text similarity metrics.

6.4 Ablation Study: Effect of Prompt Variation on Lexical Overfitting

(To evaluate whether prompt variation mitigates lexical overfitting, we conducted an ablation study comparing model performance under two settings. Due to computational cost and resource constraints, we did not regenerate the entire test set using the same prompt as the training data. Instead, we split the original training dataset and constructed a held-out validation set of 150 images using the same prompt (Prompt A). The remaining data was used to fine-tune both BLIP-2 and Lingshu models. This validation set, not seen during training, was then used as a proxy test set to evaluate the effect of prompt similarity.)*Deepali-edit*

(We compare two settings: (i) evaluation on the validation set generated using the same prompt (✓), and (ii) evaluation on the original test set generated using a different prompt (✗).)*Deepali-edit*

(As shown in Table 10, both models exhibit significantly inflated lexical scores under the same-prompt setting. For example, BLIP-2 achieves a BLEU-4 score of 0.51 and TF-IDF of 0.52 (✓), compared to 0.18 and 0.18 (✗). Similarly, Lingshu shows an increase in BLEU-4 from 0.02 to 0.10 and TF-IDF from 0.04 to 0.32. This indicates strong lexical overfitting, where models rely on prompt-specific patterns rather than true clinical reasoning.)*Deepali-edit*

(In contrast, BERTScore and semantic similarity (BioClinicalBERT) remain relatively stable across both settings, highlighting their reduced sensitivity to lexical overlap. These findings provide empirical evidence that constructing training and evaluation data using distinct prompts is essential to prevent lexical overfitting and ensure that model performance reflects genuine clinical reasoning rather than prompt-specific pattern exploitation.)*Deepali-edit*

Table 10. Ablation study comparing fine-tuned model performance under same-prompt (✓) and different-prompt (✗) evaluation. Same-prompt results use a held-out validation set generated with Prompt A, while different-prompt results use the original test set generated with Prompt B. Elevated lexical scores in the same-prompt setting indicate prompt-induced lexical overfitting.

Model Name	Same (Prompt)	BERTScore-F1	Semantic Similarity (BioClinicalBERT)	ROUGE-L	METEOR	BLEU-4	TF-IDF
BLIP2 [22]	✗	0.5472	0.9432	0.4109	0.3465	0.1809	0.1803
BLIP2 [22]	✓	0.7796	0.9623	0.3184	0.5670	0.5156	0.5200
Lingshu [40]	✗	0.5463	0.8981	0.1562	0.1628	0.0221	0.0373
Lingshu [40]	✓	0.6516	0.9375	0.3077	0.3190	0.1085	0.3162

6.5 Fine-Tuning Configurations

(Table 11 summarizes the hyperparameters used during fine-tuning of all baseline models, including learning rate, per-device batch size, number of epochs, gradient accumulation steps, and model parameter size. Models are categorized into (1) *full fine-tuning*, where all parameters are updated end-to-end, and (2) *LoRA/QLoRA fine-tuning*, where only low-rank adapter modules are optimized for parameter-efficient training.)*Deepali-edit*

(All experiments were conducted on a workstation equipped with two NVIDIA A6000 GPUs (48 GB VRAM each), ensuring consistent computational settings. Input images were resized to model-specific resolutions (typically 224x224) and normalized using standard ImageNet statistics, without incorporating annotation overlays. A unified instruction-style prompting scheme was adopted across models,

without enforcing chain-of-thought reasoning, and outputs were constrained to concise clinical answers. During inference, greedy decoding was used with temperature = 0.0, top-p = 1.0, and a maximum generation length of 256 tokens.)*Deepali-edit*

Table 11. Fine-tuning configurations for all **SpineXR-VQA** baseline models. Batch size denotes per-device batch size; gradient accumulation indicates the number of update steps accumulated before each parameter update.

Model	Params	LR	Batch	Epochs	Grad. Accum.	LoRA
Full Fine-Tuning Models						
LLaVA-Phi-3 Mini	3.8B	5×10^{-5}	4	5	4	No
BLIP-2 (FLAN-T5-XL)	3B+	5×10^{-5}	2	5	4	No
Qwen2.5-VL-3B-Instruct	3B	4×10^{-5}	4	5	2	No
LLaVA-Med v1.5 (Mistral-7B)	7B	5×10^{-5}	1	3	8	No
LoRA / QLoRA Fine-Tuning Models						
MedGemma-4B	4B	1×10^{-4}	1	3	16	Yes
MedFlamingo (9B + 7B)	9B+	5×10^{-5}	1	3	4	Yes
Lingshu-7B	7B	5×10^{-5}	1	3	8	Yes

7 Discussion and Future Directions

The construction and analysis of **SpineXR-VQA** provide insight into the practical and methodological challenges of developing clinically grounded medical VQA benchmarks. Rather than emphasizing dataset constraints, this section interprets the empirical findings observed during dataset validation and model analysis, draws broader insights about current vision-language modeling capabilities in spine imaging, and outlines implications for future dataset design and evaluation.

- **Interpretation of empirical findings.** Our error analysis reveals a consistent asymmetry in the performance of current vision-language models within the context of **SpineXR-VQA**: models are substantially more reliable at generating syntactically and semantically well-formed questions than clinically accurate answers. For LLaMA 3.2 Vision-Instruct, question correctness exceeded 85% across validation subsets, while over one-third of generated answers required expert correction and a further 30–40% were judged clinically vague despite linguistic plausibility. This discrepancy indicates that, although contemporary VLMs capture coarse anatomical context from spine radiographs, they struggle to translate visual cues into diagnostically grounded conclusions. Importantly, these errors were not dominated by gross hallucinations but by subtle omissions, hedging, or under-specified clinical reasoning, suggesting partial but incomplete internalization of radiographic decision-making.
- **Insights from cross-institutional expert validation.** Analysis of the 283 shared radiographs (994 QA pairs) reviewed by clinicians from both institutions revealed high agreement in question formulation but greater variability at the answer level. Rather than reflecting inconsistency in clinical judgment, this pattern highlights differences in diagnostic emphasis, reporting granularity, and terminology across expert reviewers. These findings suggest that answer-level disagreement can serve as a useful signal for identifying clinically ambiguous cases or under-specified questions. Within **SpineXR-VQA**, multi-expert review therefore functioned not only as a quality control mechanism but also as a lens for revealing inherently uncertain or borderline diagnostic scenarios.

- **Insights into dataset design trade-offs.** The empirical behavior of models evaluated on **SpineXR-VQA** reinforces the value of prioritizing clinician-verified annotations over unconstrained scale. While large synthetic datasets enable broad coverage, our analyses demonstrate that fine-grained diagnostic errors and reasoning gaps become visible only under rigorous expert scrutiny. This suggests that moderately sized, high-fidelity datasets play a complementary role to large-scale resources by exposing failure modes that are otherwise obscured by surface-level correctness. From this perspective, **SpineXR-VQA** is best understood as a diagnostic benchmark rather than a pretraining corpus.
- **Implications for future QA generation pipelines.** The observed gap between linguistic fluency and clinical correctness indicates that future QA generation pipelines may benefit from tighter integration of medical structure. Hybrid approaches that combine VLM generation with rule-based validation, anatomy-aware constraints, or post-generation expert triaging may improve answer fidelity without sacrificing scalability. Additionally, selectively allocating expert review to cases exhibiting higher model uncertainty or cross-expert disagreement could enable more efficient use of clinical resources in future dataset expansions.
- **Implications for evaluation methodology.** Our analyses further demonstrate that conventional text-similarity metrics are insufficient for characterizing clinical correctness in medical VQA. Linguistically fluent yet diagnostically incomplete answers frequently achieve high lexical similarity scores while failing clinical validation. This underscores the need for evaluation protocols that account for diagnostic adequacy, uncertainty expression, and reasoning completeness. Incorporating structured annotations, severity grading, or uncertainty labels into future iterations of **SpineXR-VQA** could support more clinically meaningful model evaluation and comparison.
- **Future directions.** Although the current release of **SpineXR-VQA** focuses on spine radiographs, the framework naturally extends to complementary imaging modalities such as CT or MRI, enabling richer multimodal reasoning across anatomical and pathological dimensions. Moreover, introducing multi-turn or contrastive questioning paradigms may better reflect real-world diagnostic workflows and encourage models to engage in explicit reasoning rather than single-step prediction.

In summary, the findings derived from **SpineXR-VQA** highlight a persistent gap between visual-language fluency and clinically grounded reasoning in contemporary multimodal models. By emphasizing expert verification, interpretability, and diagnostic relevance, **SpineXR-VQA** offers a foundation for both probing current model limitations and guiding the development of more clinically faithful medical VQA systems.

8 Limitations

While **SpineXR-VQA** provides a clinically verified benchmark for spine X-ray-based visual question answering, several limitations remain.

- (1) **Imaging modality restriction.** The dataset is limited to X-rays and does not include cross-sectional imaging modalities such as CT or MRI. Consequently, the scope of reasoning is primarily confined to osseous alignment, degenerative changes, and structural abnormalities, and does not capture soft-tissue, neurological, or vascular pathology commonly assessed in spine imaging workflows.
- (2) **Dependence on semi-automatic QA generation.** Initial question-answer pairs were generated using the LLaMA 3.2 Vision-Instruct model prior to expert review. Although all annotations were clinically validated and corrected, this process may introduce residual linguistic regularities or conceptual biases originating from the underlying generative model that are not fully eliminated through post hoc verification.

- (3) **Limited geographic diversity of expert reviewers.** Clinical validation was performed by ten experts from two countries. While this enabled cross-institutional comparison, diagnostic conventions, terminology, and reporting styles may differ in other healthcare systems, potentially limiting the generalizability of linguistic and interpretive patterns learned from the dataset.
- (4) **Dataset scale.** The dataset comprises 8,272 QA pairs derived from 2,187 unique x-rays. Although sufficient for benchmarking and error analysis, this scale may be inadequate for large-scale pretraining or for training highly data-intensive multimodal models without additional augmentation.
- (5) **Absence of patient-level clinical context.** To ensure privacy and ethical compliance, the dataset does not include patient demographics, clinical history, or laboratory findings. The absence of such contextual information limits the evaluation of context-aware diagnostic reasoning that often informs real-world clinical decision-making.
- (6) **Benchmarking coverage.** The experimental evaluation focuses on a representative subset of general-purpose open and medical vision–language models. While sufficient for characterizing baseline behavior and common failure modes, the results may not generalize to all existing or future architectures, particularly highly specialized or task-adapted models.
- (7) **Use of publicly available imaging data.** The radiographs were sourced from an existing public dataset rather than being collected directly from the institutions involved in clinical validation. As a result, image acquisition protocols may not fully align with the validating institutions’ clinical workflows, which could influence model behavior under institution-specific conditions.
- (8) **Evaluation Metrics.** (A limitation of standard evaluation metrics is their inability to explicitly penalize fine-grained anatomical errors. For instance, a one-level deviation in spinal localization (e.g., L3 vs. L4) constitutes a clinically significant error but may still receive high semantic similarity scores. To address this limitation, our structured error analysis (Appendix H) explicitly annotates anatomical mislocalization as a distinct failure mode, enabling a more clinically grounded assessment beyond aggregate metrics. While existing benchmarks in medical vision–language tasks primarily rely on lexical and embedding-based metrics, the development of location-aware evaluation metrics that explicitly weight anatomical correctness remains an important direction for future work.)*Deepali–edit* (R2-commnet2)*Deepali–Comment*

Despite these limitations, **SpineXR-VQA** serves as a rigorously validated reference point for spine radiograph-based visual question answering and provides a foundation for future extensions toward larger-scale, multi-modal, and multi-institutional benchmarks. In future work, we plan to address some of the above mentioned limitations.

9 Ethical Considerations

(The proposed SpineXR-VQA dataset is intended strictly for research and benchmarking purposes and is not designed for clinical deployment or real-world medical decision-making. In particular, the inclusion of treatment-related questions is aimed at evaluating model reasoning capabilities and should not be interpreted as clinical advice.)*Deepali–edit*

(Our analysis reveals that current multimodal models frequently produce clinically incorrect or unsafe outputs, including missed abnormalities, incorrect anatomical localization, and inappropriate management recommendations. Such failure modes, especially in treatment-related responses, pose significant risks if these systems are used without expert supervision.)*Deepali–edit*

(We emphasize that the dataset and associated models should be used solely in controlled research settings. Any potential clinical application would require rigorous validation, regulatory approval, and direct oversight by qualified medical professionals. These findings highlight the importance of

developing clinically grounded evaluation protocols and safety-aware model design in medical visual question answering.)*Deepali–edit* (R3-comment4)*Deepali–Comment*

10 Conclusion

In this work, we introduced **SpineXR-VQA**, the first fully clinician-validated, generative VQA benchmark tailored to spinal radiography. By leveraging a multi-institutional annotation workflow, we demonstrated that structured pipelines can produce linguistically diverse and diagnostically consistent datasets, effectively bridging the gap left by existing template-based or synthetic resources.

Our extensive benchmarking of fifteen state-of-the-art Multimodal Large Language Models (MLLMs) reveals a critical disparity: despite proficiency in general-domain tasks, current models exhibit significant deficits in clinical faithfulness, visual grounding, and fine-grained pathology recognition. These findings underscore the inadequacy of relying solely on lexical metrics and highlight the urgent need for anatomy-specific domain adaptation and tailored evaluation metric.

While this study is bounded by its focus on 2D radiography and semi-automated generation, these constraints clearly define the trajectory for future research. Subsequent efforts must prioritize incorporating different modalities (CT and MRI), integrating multi-turn reasoning, and employing active-learning pipelines to scale expert verification. Ultimately, **SpineXR-VQA** establishes a rigorous foundation for advancing generative SpineVQA.

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A Prompt Templates for QA Generation

To ensure that generated question–answer pairs reflected clinically meaningful reasoning rather than templated or generic descriptions, we designed two structured prompt templates and used them with the **LLaMA 3.2 11B Vision-Instruct** model for semi-automatic QA generation. **Prompt A** was used for the creation of the **training set**, emphasizing diversity and balanced coverage across six reasoning categories. **Prompt B** was used for the **test set**, emphasizing open-ended, explanation-driven questions to better evaluate generalization and diagnostic reasoning capabilities.

Both prompts constrained the LLM to produce concise, factual, anatomy-grounded answers without speculation or references to unavailable clinical history. The full prompt templates are shown below for reproducibility.

PROMT A: Train Set Generation

```

messages = [
{
  "role": "user",
  "content": [
    {
      "type": "text",
      "text": ""
    }
    You are a medical AI assistant analyzing an X-ray image. Your task is to examine the image and generate structured question-answer pairs.
    Analyze the image, and if there is no annotation, it means there is no abnormality in the image. If there are abnormalities, generate six explicitly written
    questions and their respective answers, considering all detected abnormalities together. The categories are:

    1. Abnormality: Write a question about the abnormalities and provide a detailed response. The question can start with "What," "Is," or "Are."
    2. Severity: Write a question about the severity of the abnormalities and provide an answer. The question can start with "How severe," "Is," or "Are."
    3. Location: Write a question about the location of the abnormalities and describe their placement in the image. The question can start with "Where," "Is," or "Are."
    4. Diagnostic: Write a question asking for the most likely diagnosis and provide a single diagnostic explanation. The question can start with "What is," "Is," or "Are."
    5. Reasoning: Write a question asking for the possible causes of the abnormalities and provide a response. The question can start with "What are," "Is," or "Are."
    6. Treatment: Write a question about the suggested treatment or management options for the detected abnormalities and provide a relevant answer. The question
    can start with "What treatment," "Is," or "Are."

    If there is no abnormality detected, generate only one question about abnormality, such as ""Is there any abnormality in the image?"" and provide the answer:
    ""No abnormality detected."" Do not generate any additional questions.

    Rules:
    - Each question must be explicitly written, followed by its corresponding answer.
    - Format the response clearly with each question first, followed by its answer.
    - If there are multiple abnormalities, address them together in a single response instead of separate question sets.
    - Do not repeat the question in the answer.
    - Do not copy these instructions in your response.
    - Use the exact names of the detected abnormalities in the answers.

    ""
  ],
  {
    "type": "image_url",
    "image_url": {
      "url": "f"data:image/jpeg;base64,{base64_image}"
    }
  }
]

```

Fig. 9. Prompt A used for generating training-set QA pairs with **LLaMA 3.2 Vision-Instruct**. This template directs the model to produce six QA pairs per X-ray image, covering abnormality, location, severity, diagnosis, reasoning, and treatment.

B Dataset Structure and Data Fields

Upon acceptance, the cleaned and verified dataset will be released via Hugging Face. The dataset is organized into two main components: (1) CSV files and (2) corresponding image folders.

PROMPT B: Test Set Generation

```

messages = [
  {
    "role": "user",
    "content": [
      {
        "type": "text",
        "text": f"""
You are a medical AI assistant analyzing an X-ray image. Your task is to examine the image and generate structured question-answer pairs.
Analyze the given image ({image_id}), and if there is no annotation, that means there is no abnormality in the image.
If there are abnormalities, generate six explicitly written questions and their respective answers, considering all detected abnormalities together. The
categories are:

1. Abnormality: Write a question about the abnormalities and provide a detailed response.
2. Severity: Write a question about the severity of the abnormalities and provide an answer.
3. Location: Write a question about the location of the abnormalities and describe their placement in the image.
4. Diagnostic: Write a question asking for the most likely diagnosis and provide a single diagnostic explanation.
5. Reasoning: Write a question asking for the possible causes of the abnormalities and provide a response.
6. Treatment: Write a question about the suggested treatment or management options for the detected abnormalities and provide a relevant answer.

If there is no abnormality detected, generate only one question-answer pair stating:
- Question: "Is there any abnormality present in this image?"
- Answer: "No abnormality detected."

Rules:

Avoid starting any question with the following words: what, where, which, is, or are.

Use diverse, professional clinical phrasing that maintains clarity and avoids redundancy.

Ensure each question is explicitly written and followed by its corresponding answer.

Do not repeat the question's wording in the answer.

Avoid including question labels like "Question 1: Abnormality."

Keep the language professional, concise, and structured.
"""
      },
      {
        "type": "image_url",
        "image_url": {
          "url": f"data:image/jpeg;base64,{base64_image}"
        }
      }
    ]
  }
]

```

Fig. 10. Prompt B used for generating test-set QA pairs emphasizing reasoning depth and open-ended interpretive questions. This prompt is designed to test the generalization ability of multimodal models beyond templated pattern recognition.

B.1 CSV Files (Final_csv)

The Final_csv directory contains four files corresponding to dataset splits: India_train, India_test, Thai_train, and Thai_test. Each CSV file contains the following fields as shown in Table 12

B.2 Image Folders (Final_images)

The Final_images directory contains image data organized by split: India_train, India_test, Thai_train, and Thai_test.

Each split includes:

- **annotated_images/**: PNG images with clinical annotations
- **unannotated_images/**: Raw PNG images without annotations

Field Name	Description	Example
image_id	Unique image identifier (file name)	Vindr_031.png
category	Question category (abnormality, severity, location, treatment, reasoning, diagnostic)	abnormality
question	Natural language question	What abnormality is present?
answer	Ground truth answer	Fracture at L1–L4

Table 12. Description of dataset fields.

C Train and Test Set Examples

Tables 13 and 14 present representative question–answer examples from the training and test splits of **SpineXR-VQA**. The examples illustrate the diversity of clinical reasoning captured by the dataset, ranging from abnormality detection in normal radiographs to multi-step interpretation involving severity assessment, anatomical localization, diagnostic inference, and treatment planning in abnormal cases. Notably, the test-set examples demonstrate greater linguistic and clinical variability, reflecting the use of distinct prompt templates for train and test splits and reinforcing the dataset’s focus on evaluating generalization rather than memorization.

D Clinician Validation Form

To evaluate the SpineXR-VQA dataset, we developed a structured Google Form interface that enables experts to systematically assess generated question–answer pairs alongside their corresponding spine radiographs (Figure 11). For each image, annotators first evaluate the clinical appropriateness of the question and subsequently assess the correctness of the answer based on visual evidence. To support high-quality annotation, the interface provides a *Not Sure* option for ambiguous cases and free-text fields for annotators to supply corrected questions or answers when errors are identified. This uniform validation framework, applied across all six QA categories, facilitates both quantitative analysis (e.g., inter-annotator agreement) and qualitative error inspection.

E Fine-Grained Comparison of LLaMA and Gemini QA Outputs

To examine semantic fidelity and lexical variation between model-generated QA pairs, Table 15 presents a detailed side-by-side comparison of **LLaMA 3.2 Vision-Instruct** and **Gemini-Pro Vision** outputs across two representative **SpineXR-VQA** cases. Inline highlights mark semantically aligned phrases across models, illustrating consistency in clinical interpretations despite lexical diversity.

Table 15. Fine-grained comparison of LLaMA and Gemini QA pairs for two representative **SpineXR-VQA** cases. Inline highlights mark semantically similar phrases across models.

QA Type	LLaMA Question	LLaMA Answer	Gemini Question	Gemini Answer
Example 1: Image vindr_train_015.png (cervical disc space narrowing)				
Abnormality	What is the nature of the abnormality observed in the X-ray image?	The X-ray image shows disc space narrowing at the C5–C6 and C6–C7 levels, indicating degenerative change in the intervertebral discs.	What abnormalities are visible in this cervical spine X-ray?	The X-ray reveals disc space narrowing at two adjacent levels and the formation of osteophytes (bone spurs).

Continued on next page

QA Type	LLaMA Question	LLaMA Answer	Gemini Question	Gemini Answer
Severity	What is the severity of the disc space narrowing observed in the X-ray image?	The disc space narrowing appears to be moderate, with a noticeable reduction in the height of the intervertebral discs at the affected levels.	How severe do the disc space narrowing and osteophytes appear?	The findings suggest moderate degenerative changes, given the clear reduction in disc height and the visible presence of osteophytes.
Location	Where are the disc space narrowing observed in the X-ray image?	The disc space narrowing is observed at the C5–C6 and C6–C7 levels, which are located in the cervical spine.	Where are the disc space narrowing and osteophytes located?	These abnormalities are located in the lower cervical spine, specifically at the C5–C6 level (disc space narrowing and osteophytes) and the C6–C7 level (disc space narrowing).
Diagnostic	What is the most likely diagnosis based on the X-ray image?	The most likely diagnosis is degenerative disc disease, which is characterized by the progressive deterioration of the intervertebral discs.	What is the most likely diagnosis based on these findings?	The combination of disc space narrowing and osteophytes is characteristic of cervical spondylosis, also known as degenerative disc disease or arthritis of the neck.
Reasoning	What are the possible causes of the disc space narrowing observed in the X-ray image?	The possible causes of the disc space narrowing include age-related wear and tear, trauma, or underlying medical conditions such as osteoarthritis or spinal stenosis.	What are the possible causes of disc space narrowing and osteophytes?	These changes are typically caused by age-related wear and tear (degeneration) of the intervertebral discs and joints; as discs lose height, the vertebrae get closer, leading to disc space narrowing and the formation of osteophytes.
Example 2: Image vindr_train_294.png (multi-level lumbar degeneration)				
Abnormality	What are the abnormalities detected in the X-ray image?	The X-ray image shows multiple osteophytes, disc space narrowing, and other lesions.	What abnormalities are identified in this X-ray of the lumbar spine?	The image reveals multiple abnormalities, including extensive osteophytes, disc space narrowing, and other lesions.

Continued on next page

QA Type	LLaMA Question	LLaMA Answer	Gemini Question	Gemini Answer
Severity	What is the severity of the abnormalities detected in the X-ray image?	The severity of the abnormalities is moderate to severe, with multiple osteophytes and disc space narrowing indicating significant degenerative changes.	How severe do these combined findings appear?	The presence of widespread, multi-level osteophytes along with disc space narrowing and other lesions indicates significant and advanced degenerative changes in the spine.
Location	Where are the abnormalities located in the X-ray image?	The abnormalities are located in the lumbar spine, specifically in the L2–L5 vertebrae, with multiple osteophytes and disc space narrowing observed.	Where are these abnormalities located?	The osteophytes are present on the vertebral bodies throughout the entire lumbar spine from L1 to L5; disc space narrowing and other lesions are specifically noted around the L2–L3 level.
Diagnostic	What is the most likely diagnosis based on the X-ray image?	The most likely diagnosis is degenerative disc disease, with multiple osteophytes and disc space narrowing indicating significant degenerative changes.	What is the most likely diagnosis based on these findings?	The combination of extensive osteophytes and disc space narrowing is highly characteristic of advanced lumbar spondylosis (degenerative osteoarthritis of the spine); the other lesions are likely part of this severe degenerative process.
Reasoning	What are the possible causes of the abnormalities detected in the X-ray image?	The possible causes of the abnormalities are age-related degeneration, trauma, or genetic predisposition.	What are the likely causes of such widespread osteophytes, disc space narrowing, and other lesions?	These findings are typically the result of chronic, age-related wear and tear on the intervertebral discs and facet joints, leading to loss of disc height, joint instability, and compensatory osteophyte formation.
Treatment	What are the suggested treatment or management options for the detected abnormalities?	The suggested treatment or management options are conservative measures such as physical therapy, pain management, and lifestyle modifications, with possible surgical intervention for severe cases.	What treatment or management options might be suggested for these conditions?	Management usually starts with conservative care, including physical therapy, pain relievers (NSAIDs), and activity modification; if pain is severe or there is nerve compression, injections or surgical intervention may be considered.

Section 51: vindr_test_1059.png

Description (optional)

Image 51: vindr_test_1059.png



[vindr_test_1059.png] Q1 (Abnormality): Can the X-ray image reveal any discernible abnormalities?

Answer: The X-ray image shows a significant loss of vertebral body height, indicating a compression fracture.

Description (optional)

[vindr_test_1059.png] Q1 - Is the above question appropriate with respect to the image? *

☐ Question correct

☐ Question not correct

☐ Not Sure

[vindr_test_1059.png] Q1 - Please write the correct question about Abnormality in the image.

Long answer text

[vindr_test_1059.png] Q1 - Please write the correct question about Abnormality in the image.

Long answer text

[vindr_test_1059.png] Q1 - Is the answer given to the above question correct? *

☐ Answer correct

☐ Answer not correct

☐ Not Sure

[vindr_test_1059.png] Q1 - If the answer is inappropriate, please provide the correct answer.

Long answer text

[vindr_test_1059.png] Q2 (Severity): Assess the degree of displacement and comminution in the compression fracture.

Answer: The compression fracture appears to be severe, with a significant loss of vertebral body height, suggesting a high degree of instability in the spine.

Description (optional)

[vindr_test_1059.png] Q2 - Is the above question appropriate with respect to the image? *

☐ Question correct

☐ Question not correct

☐ Not Sure

[vindr_test_1059.png] Q2 - Please write the correct question about Severity in the image.

Long answer text

[vindr_test_1059.png] Q2 - Is the answer given to the above question correct? *

☐ Answer correct

Fig. 11. Excerpt of the Google Form used for clinical validation. Each form presents a spine X-ray image along with generated question-answer pairs. Clinicians assess the correctness of each question and answer using structured options (*Correct*, *Not Correct*, *Not Sure*) and provide corrections when necessary.

F Sample Examples of all the Models

In this section, we present two representative qualitative outputs for each evaluated model, considering both the zero-shot and fine-tuned settings. Specifically, one sample corresponds to a normal spinal X-ray (Image vindr_test_101.png), while the other represents an abnormal case exhibiting clinically relevant pathology (Image vindr_test_1005.png). For each image, we report model responses to the same set of questions, enabling a consistent cross-model comparison. The consolidated outputs for all models are summarized in Table 16. Our analysis reveals that several zero-shot models tend to generate overly generic responses, with some incorrectly interpreting spinal radiographs as chest X-rays or brain images. Additionally, certain models misclassified normal images as abnormal, indicating limited domain awareness and poor anatomical grounding in the zero-shot setting. In contrast, fine-tuned models demonstrate improved anatomical specificity and clinical relevance, particularly in distinguishing normal from pathological findings. These examples highlight the limitations of

zero-shot general-purpose models for spine-specific VQA tasks and underscore the importance of domain-adapted fine-tuning.

Table 16. Representative sample example from all the models. The image identifier is provided as an example header to reduce column redundancy.

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Proprietary Models (Zeroshot)			
Model: Claude Sonnet			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The X-ray shows a dilated bowel loop in the right upper quadrant, suggesting possible bowel obstruction or ileus.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Mild osteophyte formation is present along the lumbar vertebral bodies, consistent with degenerative changes.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes typically develop at the vertebral body margins, particularly in the lumbar and cervical spine regions.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Based on the X-ray image, the most probable clinical interpretation is a normal spine with no acute abnormalities detected.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes in the spine typically develop due to degenerative changes, chronic mechanical stress, or osteoarthritis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes typically involve conservative management with physical therapy, NSAIDs, and, in severe cases, surgical intervention to remove the bony outgrowths.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	I am unsure based on this image.

Continued on next page

Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Model: GPT-4.1			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	There is no obvious radiographic abnormality in the spine or pelvis visible on this X-ray image.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	There is minimal to no significant osteophyte formation visible in the spine on this X-ray.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes typically develop in the cervical, thoracic, and lumbar regions of the spine, most commonly at the vertebral endplates and facet joints.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The X-ray shows a normal lumbosacral spine with no obvious fractures, dislocations, or significant degenerative changes.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Underlying factors contributing to the development of osteophytes in the spine include aging, degenerative disc disease, osteoarthritis, chronic mechanical stress, and previous spinal injury.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes typically involve physical therapy, pain management (NSAIDs), weight reduction, and in severe cases, surgical intervention.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	There are no obvious pathological features visible in this cervical spine X-ray.
Model: O4 Mini			
Example: Image vindr_test_1005.png			

Continued on next page

Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	No abnormality found
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Mild osteophyte formation.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes most commonly form at the anterior and posterior margins of the vertebral bodies in the cervical (C3–C7) and lumbar (L3–L5) regions.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Features on this abdominal X-ray are most consistent with small bowel obstruction.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Age-related intervertebral disc degeneration and cartilage loss combined with chronic mechanical stress, joint instability, and low-grade inflammation that stimulate periosteal bone formation.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Conservative measures (NSAIDs, physical therapy and activity modification), image-guided corticosteroid injections for symptom relief, and surgical osteophylectomy or decompressive procedures in refractory cases.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No, there are no obvious pathological features.
Model: Claude Haiku			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	Scoliosis.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Moderate osteophyte formation is present throughout the spine.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes typically develop in the lumbar and cervical regions of the spine.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Scoliosis.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Degenerative changes, including disc space narrowing and facet joint arthropathy.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Conservative management, including anti-inflammatory medications, physical therapy, and weight loss.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	I am unsure based on this image.
Model: GPT-5.1			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	Lumbarization of S1 (transitional lumbosacral vertebra).
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Mild anterior osteophyte formation in the lumbar spine.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes most commonly develop at the cervical (C3–C7), thoracolumbar junction (T12–L1), and lumbar (L3–L5) vertebral body margins and facet joints.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Normal lumbar spine and pelvis with no acute osseous abnormality identified.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Age-related degeneration of intervertebral discs and facet joints, mechanical stress/overload, spinal instability or malalignment, prior trauma, and underlying osteoarthritis or spondylosis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Conservative management with NSAIDs/analgesics, physical therapy and activity modification, possible corticosteroid injections, and surgical excision or decompression if severe symptoms or neurologic compromise occur.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No definite spinal or pelvic pathology is apparent on this image.
Model: GPT-5 Mini			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	Lumbarization of S1 (transitional lumbosacral vertebra).
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Mild multilevel anterior osteophyte formation consistent with mild spondylosis.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Primarily the cervical and lumbar regions—especially C5–C6 and L4–L5—with thoracic involvement less common.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Nonobstructive bowel gas pattern without radiographic evidence of acute osseous abnormality or free intraperitoneal air.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Chronic degenerative changes—especially age-related osteoarthritis and degenerative disc disease with segmental instability and abnormal mechanical loading from repetitive microtrauma or prior injury—along with chronic inflammation, metabolic conditions (e.g., DISH), obesity, and genetic predisposition.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Conservative management (NSAIDs/analgesics, physical therapy, activity modification, weight loss, and corticosteroid injections) and, if refractory or causing neural compression, surgical decompression/osteophytectomy with possible fusion.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No acute osseous pathology identified; there are only mild multilevel degenerative (spondylotic) changes of the lumbar spine.
General Purpose Open Models (Zeroshot)			
Model: Deepseek			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	There appears to be an abnormal mass or lesion located near the right side of the abdomen, which could potentially indicate a tumor or other pathological condition. Further clinical correlation would be necessary for accurate diagnosis.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The degree of osteophyte formation appears moderate based on the visible bone structures.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The sacrum, coccyx, and lumbar vertebrae (L1-L5) can all develop osteophytes.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation is that there is a mass or lesion in the left side of the abdomen, possibly indicating a tumor or other abnormal growth. Further diagnostic tests would be necessary for confirmation.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Surgical removal, physical therapy, medication, or lifestyle changes.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No pathological features are visible in the provided X-ray image.
Model: LLaVA-Phi			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	In the X-ray image, there is a radiographic abnormality in the form of a large bone spur on the left side of the pelvis. This bone spur is a bony growth that can cause discomfort or pain in the affected area. The image is in black and white, which highlights the presence of the bone spur and allows for a clear view of the skeletal structure.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Osteophytes, also known as bone spurs, typically develop in the spine at the areas where there is increased stress or degeneration. In the image, the spine is shown in black and white, and the osteophytes are visible. The most common regions where osteophytes develop are at the cervical (neck) and lumbar (lower back) spine. These spinal regions experience the most wear and tear due to daily activities and posture, leading to the formation of osteophytes as a response to the increased stress and degeneration.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes, also known as bone spurs, are bony projections that form along the edges of bones, particularly in the spine. The development of osteophytes is often associated with several underlying factors, including: 1. Aging: As people age, the spine's intervertebral discs lose their elasticity and ability to absorb shock, leading to increased stress on the vertebrae. This can result in the formation of osteophytes as the body attempts to repair the damage.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation is that there is a mass or lesion in the left side of the abdomen, possibly indicating a tumor or other abnormal growth. Further diagnostic tests would be necessary for confirmation.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes, which are bony growths that can develop around joints due to conditions like osteoarthritis, typically involve a combination of non-surgical and surgical interventions. Non-surgical interventions include: 1. Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) can help reduce pain and inflammation associated with osteophytes. 2. Physical therapy: Exercises and stretches can help improve joint mobility and strengthen the muscles surrounding the affected joint.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The image is a black and white X-ray of a person's head and neck, showing the skeletal structure. It does not reveal any pathological features, such as fractures, deformities, or abnormalities. The image focuses on the bones and does not provide any information about the soft tissues or organs in the body.
Model:Qwen 2.5			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The radiographic abnormality present in the X-ray image is a right-sided pleural effusion, which appears as a fluid-filled space surrounding the right lung field. This can be identified by the presence of a clear, fluid-filled area adjacent to the right hemithorax, which is not typically seen in normal anatomy.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The image provided is an X-ray of the lumbar spine, which shows the vertebrae and surrounding structures. To assess the degree of osteophyte formation, we would typically look for the following: 1. Vertebral Body Size: Osteophytes can cause the vertebral body to appear larger or more rounded. 2. Vertebral Body Shape: The vertebral body may have a more irregular shape due to the presence of osteophytes. 3. Spinal Canal Size: Osteophytes can narrow the spinal canal, leading to compression of the spinal cord or nerve roots. However, without a detailed analysis of the specific X-ray image, it's not possible to provide a precise assessment of the degree of osteophyte formation. If you have access to the X-ray image, you can carefully examine these features yourself or consult with a radiologist for a professional.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The X-ray image shows a radiopaque foreign body in the right lower quadrant of the abdomen, which is consistent with a gallstone. Gallstones can cause symptoms such as abdominal pain, nausea, and vomiting. The location of the stone in the right lower quadrant suggests it may be in the gallbladder or a bile duct. Further diagnostic tests, such as ultrasound or MRI, would be necessary to confirm the diagnosis and determine the appropriate treatment.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The image you provided is an X-ray of the lumbar spine, showing the vertebrae and surrounding structures. Osteophytes are bony growths that form on the edges of the vertebrae in the spine. Treatment strategies for osteophytes typically involve: 1.Pain Management: This can include nonsteroidal anti-inflammatory drugs (NSAIDs), muscle relaxants, and other pain medications to manage symptoms. 2.Physical Therapy: Exercises to strengthen the muscles around the spine and improve flexibility can help reduce pain and prevent further damage.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes, which are bony growths that can develop around joints due to conditions like osteoarthritis, typically involve a combination of non-surgical and surgical interventions. Non-surgical interventions include: 1. Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) can help reduce pain and inflammation associated with osteophytes. 2. Physical therapy: Exercises and stretches can help improve joint mobility and strengthen the muscles surrounding the affected joint.
Example: Image vindr_test_101.png			

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The image appears to be an X-ray of the neck and upper chest region, showing the bones of the spine, ribs, and part of the shoulder blades. However, without specific clinical information or a detailed examination by a radiologist, it is not possible to definitively determine if there are any pathological features present in this particular X-ray.
Model:BLIP2			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The radiographic abnormality present in the X-ray image is a right-sided pleural effusion, which appears as a fluid-filled space surrounding the right lung field. This can be identified by the presence of a clear, fluid-filled area adjacent to the right hemithorax, which is not typically seen in normal anatomy.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The image provided is an X-ray of the lumbar spine, which shows the vertebrae and surrounding structures. To assess the degree of osteophyte formation, we would typically look for the following: 1. Vertebral Body Size: Osteophytes can cause the vertebral body to appear larger or more rounded. 2. Vertebral Body Shape: The vertebral body may have a more irregular shape due to the presence of osteophytes. 3. Spinal Canal Size: Osteophytes can narrow the spinal canal, leading to compression of the spinal cord or nerve roots. However, without a detailed analysis of the specific X-ray image, it's not possible to provide a precise assessment of the degree of osteophyte formation. If you have access to the X-ray image, you can carefully examine these features yourself or consult with a radiologist for a professional.

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QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The X-ray image shows a radiopaque foreign body in the right lower quadrant of the abdomen, which is consistent with a gallstone. Gallstones can cause symptoms such as abdominal pain, nausea, and vomiting. The location of the stone in the right lower quadrant suggests it may be in the gallbladder or a bile duct. Further diagnostic tests, such as ultrasound or MRI, would be necessary to confirm the diagnosis and determine the appropriate treatment.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The image you provided is an X-ray of the lumbar spine, showing the vertebrae and surrounding structures. Osteophytes are bony growths that form on the edges of the vertebrae in the spine. Treatment strategies for osteophytes typically involve: 1.Pain Management: This can include nonsteroidal anti-inflammatory drugs (NSAIDs), muscle relaxants, and other pain medications to manage symptoms. 2.Physical Therapy: Exercises to strengthen the muscles around the spine and improve flexibility can help reduce pain and prevent further damage.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes, which are bony growths that can develop around joints due to conditions like osteoarthritis, typically involve a combination of non-surgical and surgical interventions. Non-surgical interventions include: 1. Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) can help reduce pain and inflammation associated with osteophytes. 2. Physical therapy: Exercises and stretches can help improve joint mobility and strengthen the muscles surrounding the affected joint.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The image appears to be an X-ray of the neck and upper chest region, showing the bones of the spine, ribs, and part of the shoulder blades. However, without specific clinical information or a detailed examination by a radiologist, it is not possible to definitively determine if there are any pathological features present in this particular X-ray.
Medical Models (Zeroshot)			
Model:Lingshu			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The radiographic abnormality present in the X-ray image is a right-sided pleural effusion, which appears as a fluid-filled space surrounding the right lung field. This can be identified by the presence of a clear, fluid-filled area adjacent to the right hemithorax, which is not typically seen in normal anatomy.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The image provided is an X-ray of the lumbar spine, which shows the vertebrae and surrounding structures. To assess the degree of osteophyte formation, we would typically look for the following: 1. Vertebral Body Size: Osteophytes can cause the vertebral body to appear larger or more rounded. 2. Vertebral Body Shape: The vertebral body may have a more irregular shape due to the presence of osteophytes. 3. Spinal Canal Size: Osteophytes can narrow the spinal canal, leading to compression of the spinal cord or nerve roots. However, without a detailed analysis of the specific X-ray image, it's not possible to provide a precise assessment of the degree of osteophyte formation. If you have access to the X-ray image, you can carefully examine these features yourself or consult with a radiologist for a professional.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The X-ray image shows a radiopaque foreign body in the right lower quadrant of the abdomen, which is consistent with a gallstone. Gallstones can cause symptoms such as abdominal pain, nausea, and vomiting. The location of the stone in the right lower quadrant suggests it may be in the gallbladder or a bile duct. Further diagnostic tests, such as ultrasound or MRI, would be necessary to confirm the diagnosis and determine the appropriate treatment.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The image you provided is an X-ray of the lumbar spine, showing the vertebrae and surrounding structures. Osteophytes are bony growths that form on the edges of the vertebrae in the spine. Treatment strategies for osteophytes typically involve: 1.Pain Management: This can include nonsteroidal anti-inflammatory drugs (NSAIDs), muscle relaxants, and other pain medications to manage symptoms. 2.Physical Therapy: Exercises to strengthen the muscles around the spine and improve flexibility can help reduce pain and prevent further damage.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes, which are bony growths that can develop around joints due to conditions like osteoarthritis, typically involve a combination of non-surgical and surgical interventions. Non-surgical interventions include: 1. Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) can help reduce pain and inflammation associated with osteophytes. 2. Physical therapy: Exercises and stretches can help improve joint mobility and strengthen the muscles surrounding the affected joint.
Example: Image vindr_test_101.png			

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The image appears to be an X-ray of the neck and upper chest region, showing the bones of the spine, ribs, and part of the shoulder blades. However, without specific clinical information or a detailed examination by a radiologist, it is not possible to definitively determine if there are any pathological features present in this particular X-ray.
Model:Medflamingo			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The patient has a large, irregular, and well-defined osteolytic lesion in the right iliac wing. The lesion is most likely a metastatic lesion from a primary tumor in the right iliac wing.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes are bony outgrowths that develop on the vertebral bodies and facets of the spine. They are most commonly seen in the cervical spine, but can also be seen in the thoracic and lumbar spine. Osteophytes are a normal part of the aging process and are not considered pathologic. They are most commonly seen in the cervical spine, but can also be seen in the thoracic and lumbar spine. Osteophytes are a normal part of the aging process and are not considered pathologic.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	1. Lumbar spondylolysis. 2. Lumbar spondylolysis with spondylolisthesis. 3. Lumbar spondylolysis with spondylolisthesis and spondylolytic fracture. 4. Lumbar spondylolysis with spondylolisthesis and spondylolytic fracture with spondylolytic fracture. 5. Lumbar spondylolysis with spondylolisthesis and spondylolytic fracture with spondylolytic fracture and spondylolytic fracture.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes are bony outgrowths that develop on the vertebral endplates. They are a common finding in the spine and are usually asymptomatic. They are most commonly seen in the cervical spine, but can also be seen in the thoracic and lumbar spine. Osteophytes are most commonly seen in patients with degenerative disc disease.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	1. Non-operative treatment with analgesics, physical therapy, and weight loss. 2. Surgical intervention with debridement and fusion. 3. Surgical intervention with debridement and arthrodesis. 4. Surgical intervention with debridement and arthroplasty. 5. Surgical intervention with debridement and arthroplasty with a cemented prosthesis. 6. Surgical intervention with debridement and arthroplasty with an uncemented prosthesis.
Example: Image vindr_test_101.png			

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The possible causes of the osteophytes and spondylolisthesis include age-related wear and tear, trauma, or genetic predisposition. The fracture is likely recent that the disc space narrowing, which are abnormal bone growths that can be consistent with the spinal cord. Answer: The possible causes of the compression fracture include osteoporosis, or other spinal instability. The presence of osteophytes are not the normal height, which can lead to spinal instability and potential neurological complications. The presence of osteophytes are not consistent in the spine, which shows injuries or trauma.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The most likely diagnosis is a cervical spine fracture, specifically a C6 vertebral body height. Answer: The possible causes of the fracture on the X-ray image, which are a significant loss of vertebral body height and a noticeable kyphotic deformity. Answer: The possible causes of the fracture include trauma, surgical intervention, which may have been advanced the spinal cord.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most likely diagnosis is a cervical spine fracture, specifically a C6 vertebral body fracture. Answer: The most likely diagnosis is a cervical spine fracture, specifically a C5 fracture. Answer: The most likely diagnosis is a compression fracture of the C5 vertebral body, likely due to the posterior aspect.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The possible causes of the osteophytes include age-related wear and tear, trauma, or genetic predisposition. The presence of multiple osteophytes are not significant loss of height, indicating a possible tumor of the bone structure. Answer: The possible causes of the spinal cord are consistentn the spinal cord, likely due to osteoporosis. Answer: The possible causes of the spinal cord are a significant loss of the spinal canal. The presence of multiple osteophytes in the spinal cord are a significant loss of the spinal cord. The presence of multiple osteophytes are not consistent with this
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	The possible causes of the fracture include trauma, such as a tumor or cyst, and other complications. The fracture is consistent with a notable tumor, or a pre-existing spinal instability or a pre-existing condition such as a tumor or infection. The specific and a possible surgical intervention in the anterior aspect of the spinal cord are alsoed the spinal cord.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	Answer:enerative breaks or spinal abnormality in the spine, specifically at the C5-C5, and a possible kidney stone or tumor. The presence of a metal object that supports the vertebral body is a common site for degenerative changes.
Model:Llava Med			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The radiographic abnormality present in the X-ray image is a fracture of the right sacroiliac joint.

Continued on next page

Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The image shows a significant degree of osteophyte formation in the spine. Osteophytes are bony projections that form along the edges of bones, often in response to joint degeneration or arthritis. In this case, the presence of osteophytes in the spine may indicate an underlying condition or degenerative process affecting the spine.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Yes, osteophytes typically develop in the cervical and lumbar regions of the spine.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation of the X-ray image is that it shows a normal pelvis. This means that the pelvic structures appear to be within normal limits, without any visible abnormalities or issues.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The image shows the presence of osteophytes in the spine. Osteophytes are bony outgrowths that can form along the edges of bones, often in response to joint stress or degeneration. In the context of the spine, osteophytes can develop due to various factors, such as aging, degenerative changes, spinal stenosis, or other spinal conditions. It is important to consider the patient's medical history, symptoms, and any other relevant information to determine the underlying cause of the osteophytes and provide appropriate treatment.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes may include a combination of interventions, such as medication, physical therapy, and in some cases, surgical intervention. The choice of treatment depends on the severity of the condition, the patient's overall health, and the specific location and size of the osteophytes. It is important to consult a healthcare professional for personalized advice and treatment recommendations.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The chest x-ray image appears to be normal, without any pathological features.
General Purpose Open Model (Finetuned)			
Model: BLIP-2			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The X-ray image shows a fracture in the cervical spine, specifically at the C5-C6 vertebrae. The fracture is characterized by a clear break in the bone structure, indicating a significant injury.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The degree of osteophyte formation appears to be moderate, with several prominent growths visible along the spine.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The specific regions of the spine where osteophytes typically develop include the lumbar spine, specifically at the L4-L5 and L5-S1 levels.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most likely clinical interpretation is degenerative disc disease, which is characterized by the presence of osteophytes and disc space narrowing.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The underlying factors contributing to the development of osteophytes include age-related wear and tear, genetic predisposition, and previous injuries or trauma.

Continued on next page

Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes typically involve the following interventions: Conservative management: This may include physical therapy, pain management, and lifestyle modifications to reduce stress on the affected joints. Surgical intervention: In severe cases, surgical removal of the osteophytes may be necessary to alleviate symptoms and prevent further progression.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No pathological features detected.
Model: Qwen2.5 Instruct			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The osteophytes appear to be mild to moderate in severity, as they do not cause significant spinal deformity or compression.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and the patient's age.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The possible causes of the osteophytes include age-related wear and tear, genetic predisposition, and previous spinal injuries or trauma.

Continued on next page

Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	The suggested treatment is conservative management, including physical therapy, pain medication, and lifestyle modifications to reduce stress on the spine. Surgery may be considered in severe cases.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No abnormality detected.
Model: LLaVA Phi			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	These are bony growths that can occur on the edges of joints, particularly in the spine. They are visible as small, irregularly shaped bones protruding from the vertebrae. 2. Other lesions: There may be additional abnormalities present in the image, such as tumors or infections, but these are not clearly visible.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The osteophytes appear to be moderate in size, suggesting a moderate degree of formation. However, without a clear indication of the extent of the formation, it is difficult to determine the exact degree.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The specific regions of the spine where osteophytes typically develop include the lumbar spine, particularly at the L4-L5 and L5-S1 levels. These areas are prone to degenerative changes due to the high degree of flexibility and weight-bearing function they perform. Answer: The specific regions of the spine where osteophytes typically develop include the lumbar spine, particularly at the L4-L5

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation is osteoarthritis, given the presence of osteophytes and the patient's age. Answer: The most probable clinical interpretation is osteoarthritis. Location: The osteophytes are located in the lumbar spine, specifically at the L4-L5 and L5-S1 levels. Size: The osteophytes are small to moderate in size, with the
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The underlying factors contributing to the development of osteophytes in the spine include age-related wear and tear, genetic predisposition, and previous spinal injuries or trauma. Additionally, obesity and sedentary lifestyle may also play a role in the development of osteoarthritis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	1. Pain management: Medications such as acetaminophen or ibuprofen can help manage pain associated with osteophytes. 2. Physical therapy: Strengthening exercises and flexibility training can help improve joint mobility and reduce pain. 3. Lifestyle modifications: Maintaining a healthy weight, exercising regularly, and avoiding heavy lifting can help manage symptoms. 4. Surgery
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	Yes, the image reveals a pathological feature in the form of osteophytes, which are bony growths that can occur in various parts of the body, including the spine. These growths are typically seen in older adults and can be caused by a variety of factors, including degenerative joint disease, trauma, or genetic conditions. In this case, the osteophytes are likely related to degenerative changes in the spine
Medical Model (Finetuned)			
Model: MedFlamingo			

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The abnormality is a cervical spine fracture, specifically a displaced fracture at the C6 vertebra. The fracture is visible in the cervical spine, specifically at the C5-C6 level. The fracture is visible in the anterior aspect of the vertebral body, which are displaced or a traumatic injury.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The possible causes of the osteophytes and spondylolisthesis include age-related wear and tear, trauma, or genetic predisposition. The fracture is likely recent that the disc space narrowing, which are abnormal bone growths that can be consistent with the spinal cord.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The most likely diagnosis is a cervical spine fracture, specifically a C6 vertebral body height. Answer: The possible causes of the fracture on the X-ray image, which are a significant loss of vertebral body height and a noticeable kyphotic deformity.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most likely diagnosis is a cervical spine fracture, specifically a C6 vertebral body fracture. Answer: The most likely diagnosis is a cervical spine fracture, specifically a C5 fracture. Answer: The most likely diagnosis is a compression fracture of the C5 vertebral body, likely due to the posterior aspect.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The possible causes of the osteophytes include age-related wear and tear, trauma, or genetic predisposition. The presence of multiple osteophytes are not significant loss of height, indicating a possible tumor of the bone structure.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	The possible causes of the fracture include trauma, such as a tumor or cyst, and other complications. The fracture is consistent with a notable tumor, or a pre-existing spinal instability or a pre-existing condition such as a tumor or infection.

Continued on next page

Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	Enerative breaks or spinal abnormality in the spine, specifically at the C5-C5, and a possible kidney stone or tumor. The presence of a metal object that supports the vertebral body is a common site for degenerative changes.
Model: MedGemma			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The X-ray image shows a fracture of the right sacrum, characterized by a clear break in the bone structure.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The osteophytes appear to be moderately sized, indicating a moderate level of degenerative joint disease.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The specific regions of the spine where osteophytes typically develop include the lumbar spine, the thoracic spine, and the cervical spine.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation is that the patient has a fracture of the right sacrum, likely caused by a traumatic injury such as a fall or a motor vehicle accident.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The underlying factors contributing to the development of osteophytes in the spine include age-related wear and tear, genetic predisposition, and previous injuries or trauma.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	The treatment strategies for osteophytes typically involve interventions aimed at managing symptoms, slowing disease progression, and improving quality of life.
Example: Image vindr_test_101.png			

Continued on next page

Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No abnormality detected.
Model: LLaVA Med			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The X-ray image shows a fracture of the C6 vertebra, characterized by a clear break in the bone structure. Severity: The fracture appears to be a compression fracture, with the anterior aspect of the vertebral body being more compressed than the posterior aspect.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The osteophytes appear to be moderately severe, with some areas showing more pronounced growth than others. The osteophytes are also prominent, indicating advanced degenerative changes in the spine. Answer: The osteophytes are located along the edges of the vertebral bodies, particularly at the L4-L5 and L5-S1 levels.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The osteophytes appear to be moderately severe, with some areas showing more pronounced growth than others. The osteophytes are also prominent, indicating advanced degenerative changes in the spine. Answer: The osteophytes are located along the edges of the vertebral bodies, particularly at the L4-L5 and L5-S1 levels.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	he most likely diagnosis is a cervical spine fracture, specifically a C6 vertebra fracture. This diagnosis is supported by the clear visualization of the fracture on the X-ray image. The presence of a callus suggests that the fracture is in the healing process, which is consistent with a traumatic injury.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The underlying factors contributing to the development of osteophytes in the spine include age-related wear and tear, genetic predisposition, and previous injuries or trauma.

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Table 16 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	The suggested treatment or management option is conservative management, including physical therapy, pain management, and lifestyle modifications. Surgery may be considered in severe cases. Answer: The suggested treatment or management option is conservative management, including physical therapy, pain management, and lifestyle modifications. In severe cases, surgical intervention may be necessary to relieve pressure on the spinal cord and nerves	
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	Yes, there is an abnormality present in this image.

G Category-wise Evaluation Across Metrics

(We present category-wise performance across six clinical categories (*Abnormality, Severity, Location, Diagnostic, Reasoning, Treatment*) using multiple evaluation metrics. Models are grouped into *Proprietary, General-purpose*, and *Medical-specific*, with fine-tuned variants indicated using hatched patterns.)*Deepali–edit*

G.1 BERTScore

(BERTScore results indicate strong performance across all categories, with general-purpose and medical models consistently ranking among the top. Fine-tuning does not exhibit a uniform trend; while several fine-tuned models achieve top performance, others show comparable or slightly lower scores relative to their base variants. Across categories, performance is highest for *Abnormality, Severity*, and *Location*, with a mild decline observed in *Reasoning* and *Treatment*.)*Deepali–edit*

G.2 BLEU-4

(BLEU-4 scores are low across all categories, reflecting limited lexical overlap in generated responses. General-purpose models, particularly some fine-tuned variants, achieve the highest scores, while medical models show competitive performance in *Abnormality* and *Reasoning*. Proprietary models

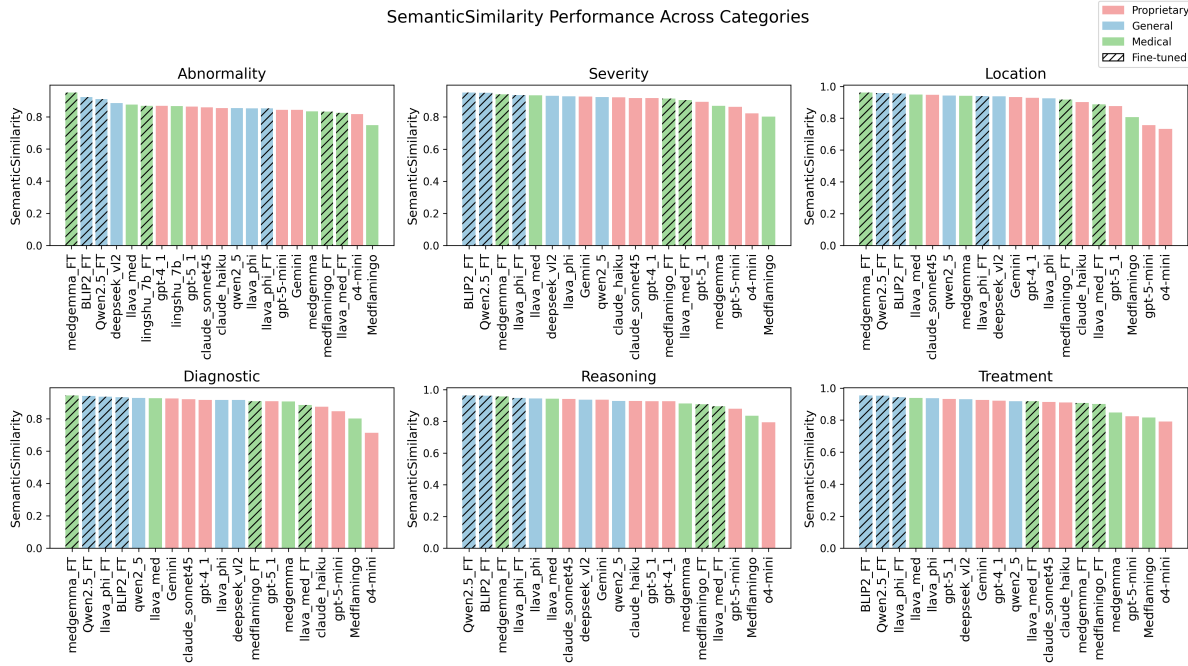


Fig. 12. Semantic similarity across clinical categories.

consistently rank lower across categories. Fine-tuning exhibits a model-dependent effect, with improvements observed in some cases but no consistent trend across all models. Performance is relatively higher in *Severity*, *Location*, and *Diagnostic*, and drops for *Reasoning* and *Treatment*.)*Deepali-edit*

G.3 ROUGE-L

(ROUGE-L shows moderate performance across categories, capturing partial structural overlap between predictions and references. General-purpose models consistently achieve the highest scores, with several fine-tuned variants among the top performers. Medical models demonstrate competitive performance, particularly in *Abnormality* and *Severity*, while proprietary models generally rank lower. The impact of fine-tuning is model-dependent, with improvements observed in some cases but no consistent trend across all models. Performance is stronger in *Severity*, *Location*, and *Diagnostic*, and declines for *Reasoning* and *Treatment*.)*Deepali-edit*

G.4 Semantic Similarity

(Semantic similarity scores are consistently high across all models and categories, indicating strong overall alignment in meaning between predictions and references. Differences between model groups are relatively small; however, general-purpose and medical models slightly outperform proprietary models in most categories. Fine-tuning shows minimal and model-dependent impact, with several fine-tuned models appearing among the top performers without a consistent trend. Performance remains uniformly strong across categories, with only slight reductions observed in *Treatment* and *Reasoning*.)*Deepali-edit*

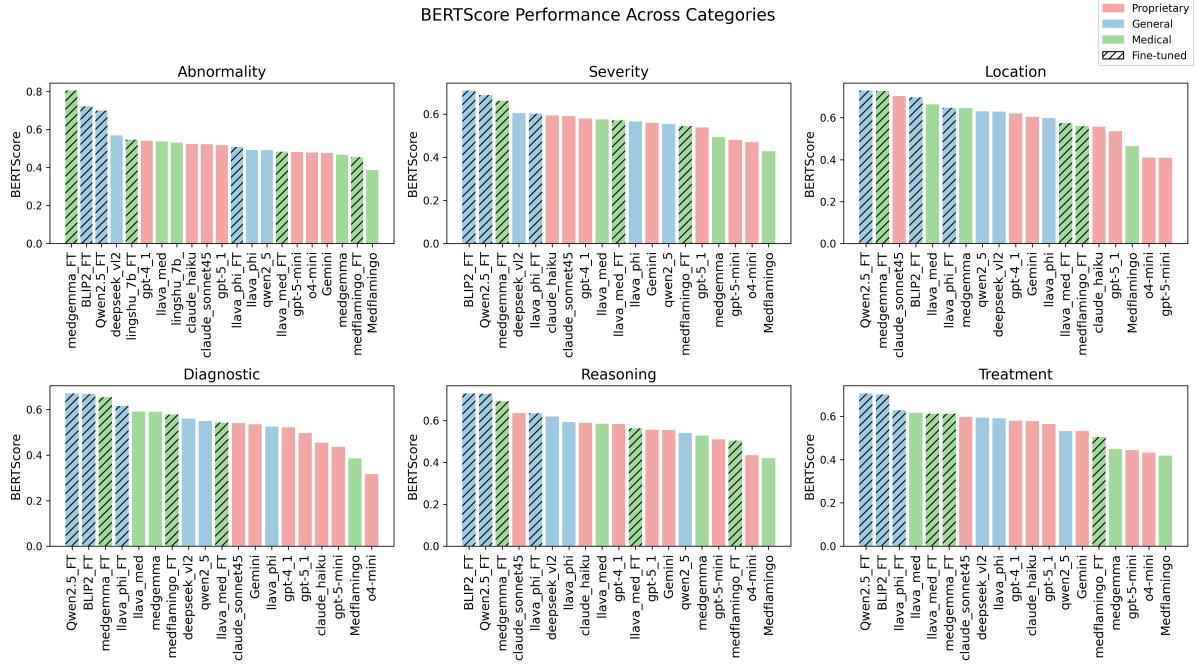


Fig. 13. BERTScore across clinical categories.

G.5 TF-IDF Similarity

(TF-IDF similarity shows moderate performance with clearer separation between model groups compared to embedding-based metrics. General-purpose and medical models consistently achieve higher scores across most categories, while proprietary models rank lower. Fine-tuned models frequently appear among the top performers, although the impact of fine-tuning remains model-dependent. Performance is relatively stronger in *Severity*, *Location*, and *Diagnostic*, and declines for *Reasoning* and *Treatment*, reflecting reduced term overlap in more complex responses.) *Deepali-edit*

H Model output error analysis

This section presents a fine-grained qualitative analysis of clinically relevant failure cases observed across the five representative models evaluated in the main text. For each model, we report selected question-answer examples covering abnormality detection, anatomical localization, diagnostic interpretation, and management reasoning, with clinically incorrect or misleading phrases explicitly highlighted. The examples reveal consistent failure patterns, including false-negative findings in the presence of fractures, substitution of acute traumatic pathology with chronic degenerative conditions, anatomically implausible localization across spinal regions, and unsafe or incomplete treatment recommendations. By directly contrasting model-generated responses with expert-validated ground truth, this analysis demonstrates how high semantic similarity scores can obscure clinically critical errors, underscoring the need for clinician-centered evaluation beyond surface-level text similarity in medical VQA systems.

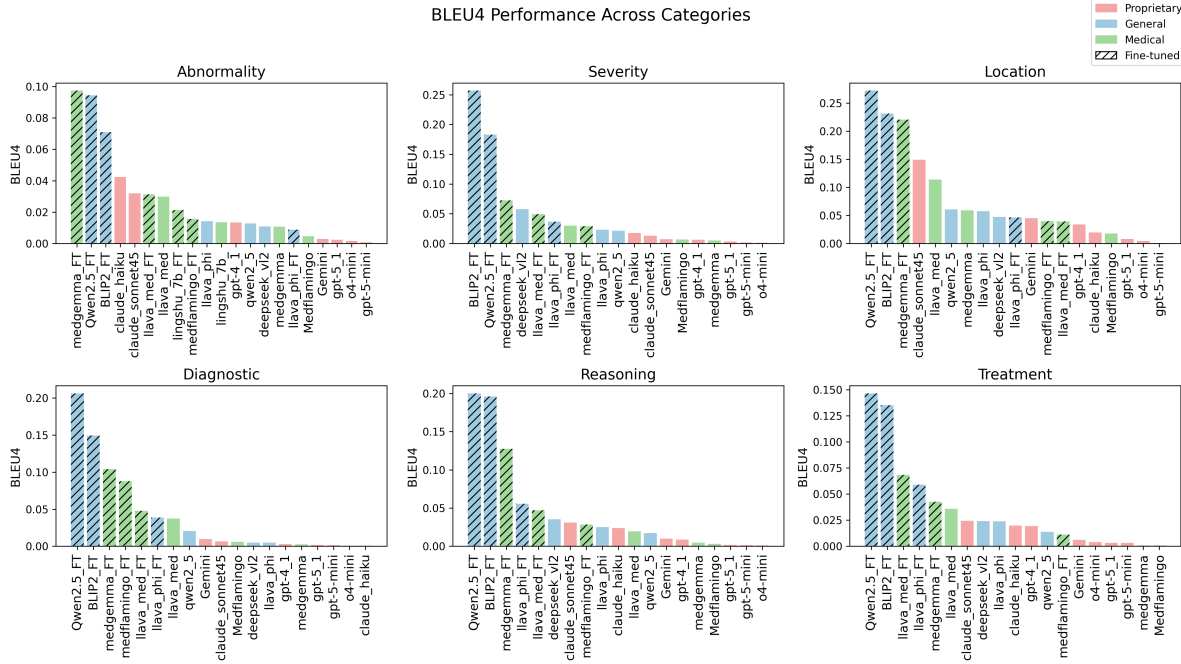


Fig. 14. BLEU-4 across clinical categories.

Table 17. Representative clinical failure cases across evaluated models. Clinically incorrect or misleading phrases are highlighted in **blue bold italics**.

Question	Ground Truth (Clinical)	Model Output	Error Type
BLIP-2			
Can you identify the radiographic abnormality present in the X-ray image?	Comminuted fracture of the C5 vertebra with vertebral body collapse and posterior element involvement.	The most likely clinical interpretation is <i>degenerative disc disease characterized by osteophytes and disc space narrowing</i> .	Pathology substitution
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are located in the lumbar spine at the L4–L5 and L5–S1 levels.	The abnormalities are located at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Does this image reveal any pathological features?	Yes. The image shows a compression fracture of the C6 vertebra.	<i>No pathological features detected</i> .	False-negative detection

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Question	Ground Truth (Clinical)	Model Output	Error Type
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar spine fracture at L1 with vertebral body displacement.	The most likely clinical interpretation is <i>degenerative disc disease with osteophyte formation</i> .	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the C6 vertebra in the cervical spine.	The anatomical site of the fracture is the <i>junction between C5 and C6</i> .	Anatomical inaccuracy
Does this image reveal any pathological features?	Yes. The image shows a fracture of the right clavicle.	<i>No pathological features detected.</i>	Missed extra-spinal pathology
Claude Sonnet			
Can you identify the radiographic abnormality present in the X-ray image?	Compression fracture of the C5 vertebra with vertebral body collapse.	The most likely clinical interpretation is <i>degenerative disc disease characterized by osteophytes</i> .	Pathology substitution
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are located in the lumbar spine at the L4–L5 and L5–S1 levels.	The abnormalities are located at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Does this image reveal any pathological features?	Yes. The image shows a compression fracture of the C6 vertebra.	<i>No pathological features detected.</i>	False-negative detection
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar spine fracture at L1 with vertebral body displacement.	The most likely clinical interpretation is <i>degenerative disc disease with osteophyte formation</i> .	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the C6 vertebra in the cervical spine.	The anatomical site of the fracture is the <i>junction between C5 and C6</i> .	Anatomical inaccuracy
Does this image reveal any pathological features?	Yes. The image shows a fracture of the right clavicle.	<i>No pathological features detected.</i>	Missed extra-spinal pathology
DeepSeek-VL2			
Can you identify the radiographic abnormality present in the X-ray image?	Compression fracture of the C6 vertebra with vertebral body collapse.	The image most likely represents <i>degenerative disc disease with osteophyte formation</i> .	Pathology substitution

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Question	Ground Truth (Clinical)	Model Output	Error Type
Does this image reveal any pathological features?	Yes. The image shows a comminuted fracture of the C5 vertebra.	<i>No pathological features detected.</i>	False-negative detection
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are present in the lumbar spine at the L1–L2 and L2–L3 levels.	The abnormalities are located at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar vertebral compression fracture with displacement.	The most likely clinical interpretation is <i>degenerative disc disease</i> .	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the L1 vertebra in the lumbar spine.	The fracture is located in the <i>cervical spine at C5–C6</i> .	Anatomical inaccuracy
Given the presence of detected abnormalities, what treatment or management strategies are recommended?	Immediate spinal immobilization with urgent surgical consultation.	<i>Conservative management with physiotherapy and lifestyle modification.</i>	Unsafe management suggestion
LingShu-7B (zero-shot)			
Can you identify the radiographic abnormality present in the X-ray image?	Comminuted fracture of the C5 vertebra with vertebral body collapse.	The findings are consistent with <i>degenerative disc disease with osteophyte formation</i> .	Pathology substitution
Does this image reveal any pathological features?	Yes. The image shows a compression fracture of the C6 vertebra.	<i>No pathological features detected.</i>	False-negative detection
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are located in the lumbar spine at the L4–L5 and L5–S1 levels.	The abnormalities are present at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar spine fracture at L1 with vertebral body displacement.	The most likely clinical interpretation is <i>degenerative disc disease</i> .	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the C6 vertebra in the cervical spine.	The fracture is located at the <i>junction between C5 and C6</i> .	Anatomical inaccuracy

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Question	Ground Truth (Clinician)	Model Output	Error Type
Given the presence of detected abnormalities, what treatment or management strategies are recommended?	Urgent spinal immobilization and surgical consultation are required.	<i>Conservative management with physiotherapy and pain control.</i>	Unsafe management recommendation
MedGemma (fine-tuned)			
Can you identify the radiographic abnormality present in the X-ray image?	Compression fracture of the C6 vertebra with loss of vertebral body height.	The findings are most consistent with <i>degenerative disc disease characterized by osteophytes.</i>	Pathology substitution
Does this image reveal any pathological features?	Yes. The image shows a fracture of the right clavicle.	<i>No pathological features detected.</i>	Missed extra-spinal pathology
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are located in the lumbar spine at the L4–L5 and L5–S1 levels.	The abnormalities are identified at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar spine fracture at L1 with vertebral body displacement.	The most likely clinical interpretation is <i>degenerative disc disease with disc space narrowing.</i>	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the L1 vertebra in the lumbar spine.	The fracture is located in the <i>cervical spine at C5–C6.</i>	Anatomical inaccuracy
Given the presence of detected abnormalities, what treatment or management strategies are recommended?	Immediate spinal immobilization with urgent surgical evaluation.	<i>Conservative treatment with physiotherapy and lifestyle modification.</i>	Unsafe management recommendation

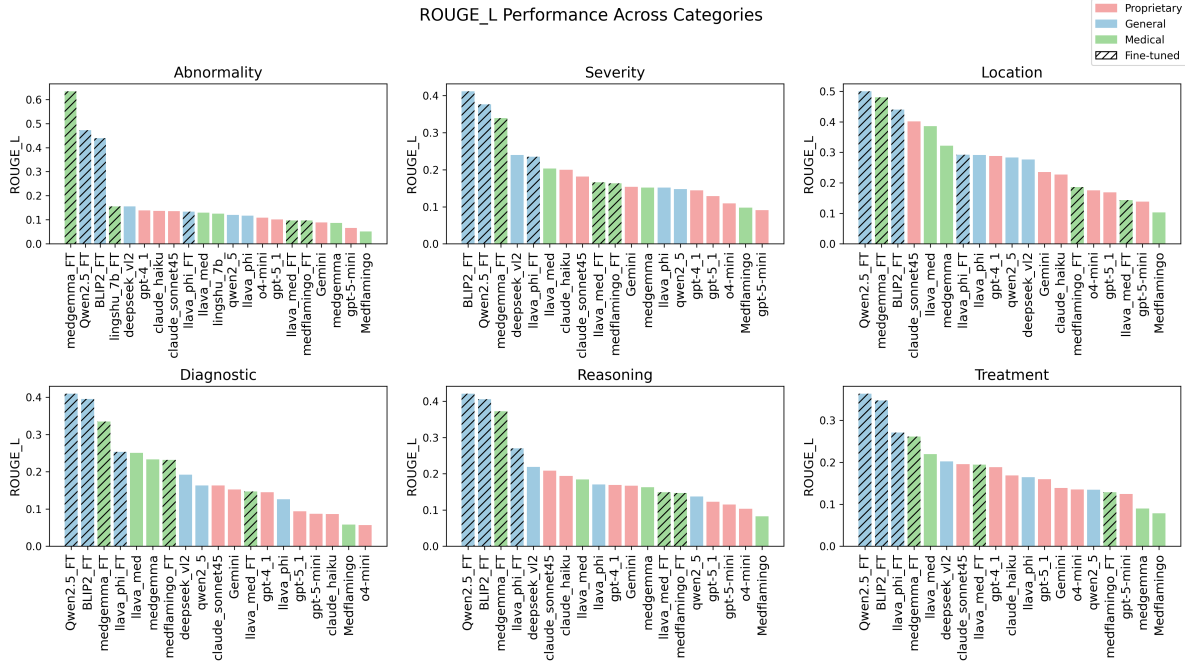


Fig. 15. ROUGE-L across clinical categories.

Table 13. Sample questions-answer pairs for Train set

Example 1: Image ID – vindr_train_4473.png		
Abnormality	Is there any abnormality present in this image?	No abnormality detected.
Example 2: Image ID – vindr_train_2027.png		
Abnormality	Is there any abnormality present in this image?	Yes, there is an abnormality present in this image. The X-ray shows a compression fracture of the L1 vertebra with evidence of callus formation, indicating a healing process.
Severity	What is the severity of the fracture?	The fracture is of moderate severity. It appears stable without significant displacement, although vertebral body compression may contribute to spinal stenosis.
Location	Where is the fracture located in the image?	The fracture is located at the L1 vertebra, just below the T12–L1 disc space, in the midline of the lumbar spine.
Diagnostic	What is the most likely diagnosis based on the X-ray image?	The most likely diagnosis is a compression fracture of the L1 vertebra, likely due to osteoporosis or trauma.
Reasoning	What are the possible causes of the fracture?	Possible causes include osteoporosis-related bone weakening or traumatic injury such as a fall or accident.
Treatment	What is the suggested treatment or management option for the detected abnormalities?	Conservative management including rest, pain control, and physical therapy is recommended, with surgical intervention considered in severe or unstable cases.
Example 3: Image ID – vindr_train_2530.png		
Abnormality	Is there any abnormality present in this image?	Yes, the image shows a compression fracture of the L1 vertebra with vertebral body height loss.
Severity	What is the severity of the fracture?	The fracture is of moderate severity and appears stable without rotational displacement.
Location	Where is the fracture located in the image?	The fracture involves the anterior aspect of the L1 vertebral body in the lumbar spine.
Diagnostic	What is the most likely diagnosis for the fracture?	A healing compression fracture of the L1 vertebra, potentially associated with osteoporosis or trauma.
Reasoning	What are the possible causes of the fracture?	Osteoporosis, traumatic injury, metabolic bone disease, or long-term medication use affecting bone density.
Treatment	What is the suggested treatment or management option for the fracture?	Conservative management including rest, analgesia, physical therapy, and possible bracing, with surgery reserved for unstable cases.

Table 14. Sample questions-answer pair for Test set

Example 1: Image ID – vindr_test_1251.png		
Abnormality	Does this image reveal any pathological features?	No abnormality detected.
Example 2: Image ID – vindr_test_538.png		
Abnormality	Can the X-ray image be evaluated for any radiographic abnormalities?	The X-ray image shows multiple abnormalities, including vertebral collapse, osteophyte formation, and the presence of a surgical implant.
Severity	Assess the degree of vertebral compression.	The vertebral collapse appears moderate to severe, with significant loss of vertebral height and potential compromise of spinal stability.
Location	Can you identify the specific areas within the image that show abnormalities?	The abnormalities are located in the lumbar spine, primarily at the L1–L2 and L2–L3 levels.
Diagnostic	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is degenerative disc disease with vertebral collapse and osteophyte formation, possibly complicated by prior surgical intervention.
Reasoning	Identify the underlying factors contributing to the observed abnormalities.	Potential contributing factors include degenerative disc disease, osteoporosis, prior trauma, or infection.
Treatment	Given the presence of detected abnormalities, what treatment or management strategies are typically recommended?	Recommended management includes conservative treatment with analgesics and physical therapy, surgical stabilization if indicated, and regular follow-up to monitor disease progression.
Example 3: Image ID – vindr_test_277.png		
Abnormality	Does this image reveal any pathological features?	Yes, the image demonstrates a compression fracture involving the C6 vertebra.
Severity	Assess the fracture's degree of displacement and comminution to determine its overall severity.	The fracture is of moderate severity, with partial compression and potential cervical spine instability but no complete displacement.
Location	Can you identify the anatomical site of the fracture depicted in the image?	The fracture is located at the C6 vertebra in the mid-cervical spine, approximately between the C5 and C7 levels.
Diagnostic	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is a compression fracture of the C6 vertebra.
Reasoning	Identify the underlying factors that may have contributed to the fracture.	Possible causes include traumatic injury, osteoporosis, or other conditions that weaken vertebral bone integrity.
Treatment	Given the presence of detected abnormalities, what are the recommended treatment or management strategies?	Management typically involves cervical spine immobilization using a collar or halo vest, followed by rehabilitation; surgical stabilization may be required in severe cases.

